

Time Series Analysis

Condensed Essentials

MATH-342 – EPFL – Prof. S. C. Olhede

Scope

The bare essentials only: **definitions**, **theorems**, **propositions**, **lemmas**, **corollaries** and the **key formulas**, chapter by chapter. No exercises, techniques, intuition boxes or pitfalls — see the full companion document for those.

Colour legend

■ Definition ■ Theorem ■ Proposition ■ Lemma/Corollary

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1 Foundations: Stationarity, Autocovariance & Ergodicity

A *time series* $\{X_t\}$ is both the data recorded in time order and the stochastic process $\{X_t : t \in T\}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ it is a realisation of. The index set T is a set of time points (here $T = \mathbb{Z}$ unless stated otherwise; we take the sampling step $\Delta = 1$). Time series differ from classical statistics because the observations are **dependent**: “there are many more ways to be dependent than to be independent”. Everything in this course is built on describing that dependence through second-order structure.

1.1 Definitions

Definition 1: (Weak / second-order) stationarity

$\{X_t\}$ is **second-order** (equivalently **weakly** or **covariance**) **stationary** if all first- and second-order joint moments exist, are finite, and are invariant to time shifts. Concretely, for all $t, s \in T$ and all admissible shifts τ :

$$(i) \mathbb{E}[X_t] = \mu; \quad (ii) \text{Var}(X_t) = \sigma^2 < \infty; \quad (iii) \mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[X_s X_{s+\tau}].$$

Condition (iii) forces $\mathbb{E}[X_t X_{t+\tau}]$ to be a function of the lag τ only.

Definition 2: Strong / strict stationarity

$\{X_t\}$ is **strictly** (completely) **stationary** if for every $n \geq 1$, every $t_1, \dots, t_n$ and every shift τ , the joint distribution of $(X_{t_1}, \dots, X_{t_n})$ equals that of $(X_{t_1+\tau}, \dots, X_{t_n+\tau})$.

Definition 3: Gaussian process

$\{X_t\}$ is a **Gaussian process** if for all n and $t_1, \dots, t_n$ the vector $(X_{t_1}, \dots, X_{t_n})$ is multivariate normal. Such a process is completely characterised by its (finite) first two moments; hence for Gaussian processes **weak** $\Leftrightarrow$ **strict stationarity**.

Definition 4: Autocovariance sequence (ACVS)

For a discrete-time second-order stationary $\{X_t\}$, the **ACVS** is

$$\gamma_\tau = \text{Cov}(X_0, X_\tau) = \text{Cov}(X_t, X_{t+\tau}), \quad \tau \in \mathbb{Z}.$$

Immediate properties: $\gamma_0 = \text{Var}(X_t) \geq 0$, symmetry $\gamma_{-\tau} = \gamma_\tau$, and (Proposition 9) positive semi-definiteness.

Definition 5: Autocorrelation sequence (ACF)

The **ACF** is $\rho_\tau = \text{Corr}(X_t, X_{t+\tau}) = \gamma_\tau / \gamma_0$, with $\rho_0 = 1$ and $|\rho_\tau| \leq 1$.

Definition 6: Sample autocovariance estimators

Given $X_1, \dots, X_n$ with sample mean $\bar{X}$, two estimators are used for $|\tau| \leq n - 1$ (both 0 otherwise):

$$\tilde{\gamma}_\tau = \frac{1}{n - |\tau|} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}), \quad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X}).$$

With known mean, $\tilde{\gamma}_\tau$ is *unbiased* but need not be positive semi-definite; $\hat{\gamma}_\tau$ is *biased* ($\mathbb{E}[\hat{\gamma}_\tau] = \frac{n-|\tau|}{n} \gamma_\tau$) but is positive semi-definite, which is why it is the default. The plug-in ACF estimator is $\hat{\rho}_\tau = \hat{\gamma}_\tau / \hat{\gamma}_0$.

Definition 7: Mean ergodicity

$\{X_t\}$ is **mean ergodic** if its first two moments are finite and $\bar{X} \xrightarrow{P} \mathbb{E}[X_t]$ as $n \rightarrow \infty$ (“temporal averages = population averages”). Ergodicity and stationarity are *not* equivalent.

1.2 Theorems & Propositions

Theorem 8: Kolmogorov existence theorem

A family of finite-dimensional distribution functions $\{F_{\mathbf{t}}\}$ are the f.d.d.’s of a stochastic process iff they are **consistent**: for every n , every $\mathbf{t}$ and every $1 \leq i \leq n$,

$$\lim_{x_i \rightarrow \infty} F_{\mathbf{t}}(\mathbf{x}) = F_{\mathbf{t}(i)}(\mathbf{x}(i)),$$

where $\mathbf{t}(i), \mathbf{x}(i)$ delete the i -th coordinate.

Why: it guarantees an infinite-index stochastic process actually exists given only its finite-dimensional “marginals”, justifying modelling $\{X_t\}_{t \in \mathbb{Z}}$ even though we observe finitely many points.

Proposition 9: The ACVS is positive semi-definite

For any n , any $t_1, \dots, t_n \in \mathbb{Z}$ and any $a_1, \dots, a_n \in \mathbb{R}$,

$$\sum_{j=1}^n \sum_{k=1}^n a_j a_k \gamma_{j-k} \geq 0.$$

Proof idea: set $W = \sum_j a_j X_j$; then $0 \leq \text{Var}(W) = \sum_{j,k} a_j a_k \text{Cov}(X_j, X_k) = \sum_{j,k} a_j a_k \gamma_{j-k}$.

Why: p.s.d. is the defining algebraic property of a valid ACVS; a candidate sequence that is *not* p.s.d. **cannot** be an autocovariance (used constantly to reject “fake” ACVS, see Exercise in Part 6).

Proposition 10: Variance of the sample mean & consistency

If $\sum_{\tau} |\gamma_{\tau}| < \infty$ then $\bar{X}$ is unbiased and

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i,j=1}^n \gamma_{j-i} = \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} \left(1 - \frac{|\tau|}{n}\right) \gamma_{\tau}, \quad n \text{Var}(\bar{X}) \xrightarrow{n \rightarrow \infty} \sum_{\tau=-\infty}^{\infty} \gamma_{\tau}.$$

Hence $\text{Var}(\bar{X}) \rightarrow 0$ and $\bar{X} \xrightarrow{P} \mu$ (consistency).

Proof idea: expand the double sum, collect by lag τ , then apply the **Cesàro summability theorem** to replace $\sum (1 - \frac{|\tau|}{n}) \gamma_{\tau}$ by $\sum \gamma_{\tau}$.

Why: this is the prototype “when can a sample average replace a population average?”. Strong positive correlation inflates $\text{Var}(\bar{X})$ (e.g. AR(1) with $\phi \rightarrow 1$): dependence, not just sample size, controls estimation accuracy.

2 ARMA Models: White Noise, MA, AR, ARMA and the PACF

The ARMA family is the workhorse of linear time series modelling. Starting from **white noise** — the indivisible “atom” of randomness — we build **moving average** (MA), **autoregressive** (AR) and combined **ARMA** processes by feeding white noise through finite linear filters and linear recursions. Two diagnostic curves run through the whole story: the autocorrelation function (ACF, ρ_{τ}) and its less-obvious partner, the **partial autocorrelation function** (PACF, α_{τ}). The punchline, encoded

in one small table, is that the ACF identifies the MA order q while the PACF identifies the AR order p . We close with the **Durbin–Levinson** recursions that compute the PACF directly from the ACF.

2.1 Definitions

Definition 1: White noise (purely random process)

$\{X_t\}$ is **white noise** (a purely random process) if it is a sequence of *uncorrelated*, mean-zero, finite-variance random variables: for all $t \in \mathbb{Z}$,

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2 < \infty,$$

and $\text{Cov}(X_s, X_t) = 0$ for $s \neq t$. Equivalently its second-order structure is

$$\gamma_\tau = \begin{cases} \sigma^2 & \tau = 0, \\ 0 & \tau \neq 0, \end{cases} \quad \rho_\tau = \begin{cases} 1 & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

White noise is automatically (weakly) stationary and is the **building block** of every model below. We write ε_t for a mean-zero white noise of variance σ_ε^2 .

Definition 2: Moving average process MA(q)

Let $\{\varepsilon_t\}$ be mean-zero white noise of variance σ_ε^2 . The q -th order moving average process MA(q) is

$$X_t = \mu - \sum_{j=0}^q \theta_j \varepsilon_{t-j}, \quad \theta_0 = -1, \quad \theta_q \neq 0.$$

(The convention $\theta_0 = -1$ makes the leading term $-\theta_0 \varepsilon_t = \varepsilon_t$.) Simple examples use the equivalent zero-mean one-sign convention $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$ for MA(1). An MA(q) is just a finite linear filter of white noise, hence always stationary; the only requirement is $|\theta_j| < \infty$.

Definition 3: Moments of an MA(q)

For the MA(q) of Definition 2, with $\theta_j := 0$ for $j > q$,

$$\mathbb{E}[X_t] = \mu, \quad \gamma_\tau = \text{Cov}(X_{t+\tau}, X_t) = \begin{cases} \sigma_\varepsilon^2 \sum_{j=0}^q \theta_j \theta_{j+|\tau|} & |\tau| \leq q, \\ 0 & |\tau| > q. \end{cases}$$

Hence the **ACVS of an MA(q) cuts off (is exactly 0) for lags $|\tau| > q$** — this is the signature used to read q off the ACF.

Definition 4: Autoregressive process AR(p)

Let $\{\varepsilon_t\}$ be mean-zero white noise. The p -th order autoregressive process AR(p) is

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \varepsilon_t, \quad \phi_p \neq 0.$$

Unlike the MA case there *are* constraints on $\{\phi_j\}$ for stationarity (the roots of the AR polynomial must lie outside the unit circle), and there is no simple closed form for the ACVS in general.

Definition 5: ARMA(p, q) process

$\{X_t\}$ is an **autoregressive moving average** process of orders p and q , ARMA(p, q), if

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k \varepsilon_{t-k}, \quad \theta_0 = -1,$$

with $\{\varepsilon_t\}$ mean-zero white noise and ϕ_j, θ_k as in the AR and MA cases. Using the backshift operator B (with $B^j X_t = X_{t-j}$) this is written compactly as

$$\Phi(B)X_t = \Theta(B)\varepsilon_t, \quad \Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j, \quad \Theta(z) = - \sum_{k=0}^q \theta_k z^k.$$

As for AR, conditions on the ϕ_j are needed for the model to be valid, and closed-form ACVS is generally unavailable (though computable).

Definition 6: Causal ARMA(p, q)

An ARMA(p, q) process is **causal** (more precisely a causal function of $\{\varepsilon_t\}$) if there exists a sequence $\{\psi_j\}$ with $\sum_{j=0}^{\infty} |\psi_j| < \infty$ such that

$$X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}, \quad t = 0, \pm 1, \pm 2, \dots$$

i.e. X_t depends only on *present and past* noise. The $\{\psi_j\}$ are the **ψ -weights** (the MA(∞) representation). A mean-zero MA(q) is *always* causal (it already has this form with finitely many nonzero ψ_j). For ARMA, causality holds iff all roots of $\Phi(z)$ lie *outside* the unit circle.

Definition 7: Best linear predictor

Given mean-zero random variables $X_1, \dots, X_n, Y$, the **best linear predictor** of Y from $X_1, \dots, X_n$ is

$$P_{X_1, \dots, X_n}(Y) = \sum_{j=1}^n \beta_j X_j,$$

where $\beta_1, \dots, \beta_n$ minimise the mean-squared error $\mathbb{E}[(Y - P_{X_1, \dots, X_n}(Y))^2]$. The minimiser is characterised by the **prediction (orthogonality) equations**: the prediction error is uncorrelated with every regressor,

$$\text{Cov}(Y - P_{X_1, \dots, X_n}(Y), X_k) = 0, \quad k = 1, \dots, n.$$

Definition 8: Partial correlation

For random variables X, Y and confounders $Z = (Z_1, \dots, Z_n)^\top$, the **partial correlation** of X and Y given Z is

$$\rho_{XY \bullet Z} = \text{Corr}(X - P_Z(X), Y - P_Z(Y)),$$

i.e. the correlation of the residuals after removing the best linear effect of Z from each. If X, Y, Z are jointly Gaussian then $\rho_{XY \bullet Z} = \mathbb{E}[\text{Corr}(X, Y \mid Z)]$, and in fact $\rho_{XY \bullet Z} \stackrel{\text{a.s.}}{=} \text{Corr}(X, Y \mid Z)$.

Definition 9: Partial autocorrelation function (PACF)

For a mean-zero stationary $\{X_t\}$, the **PACF** at lag τ is the partial correlation of X_t and $X_{t+\tau}$ given the intervening values $Z = (X_{t+1}, \dots, X_{t+\tau-1})$. Writing $\hat{X}_t = P_{X_{t+1}, \dots, X_{t+\tau-1}}(X_t)$ and

$$\hat{X}_{t+\tau} = P_{X_{t+1}, \dots, X_{t+\tau-1}}(X_{t+\tau}),$$

$$\alpha_\tau = \text{Corr}(X_{t+\tau} - \hat{X}_{t+\tau}, X_t - \hat{X}_t).$$

Equivalently (regression form): if $P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$ is the best linear predictor of X_τ from its τ predecessors, then $\alpha_\tau = \alpha_{\tau,\tau}$ (the *last* regression coefficient). Conventionally $\alpha_1 = \rho_1$.

2.2 Theorems & Propositions

Proposition 10: Causal AR(1) expansion and its ACVS

For the AR(1) $X_t = \phi X_{t-1} + \varepsilon_t$ with $|\phi| < 1$, iterating the recursion gives the causal (MA(∞)) representation

$$X_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}, \quad \psi_k = \phi^k, \quad \sum_k |\psi_k| = \frac{1}{1-|\phi|} < \infty,$$

with $\mathbb{E}[X_t] = 0$ and autocovariance

$$\gamma_\tau = \frac{\sigma_\varepsilon^2}{1-\phi^2} \phi^{|\tau|}, \quad \rho_\tau = \phi^{|\tau|}.$$

Proof idea: substitute $X_{t-1} = \phi X_{t-2} + \varepsilon_{t-1}$ repeatedly; the remainder $\phi^n X_{t-n} \rightarrow 0$ in mean square. Then $\gamma_\tau = \sigma_\varepsilon^2 \sum_{k \geq 0} \phi^k \phi^{k+|\tau|} = \sigma_\varepsilon^2 \phi^{|\tau|} \sum_{k \geq 0} \phi^{2k} = \sigma_\varepsilon^2 \phi^{|\tau|} / (1 - \phi^2)$ by the geometric series.

Why: the AR(1) is the canonical example of a causal process and the template for “write an AR as an infinite MA, then use the white-noise filter trick to get the ACVS”. The ACF $\phi^{|\tau|}$ *tails off* (never hits exactly 0), motivating the PACF.

Proposition 11: ACF–PACF relation (prediction equations)

For a mean-zero stationary $\{X_t\}$, fix $\tau > 0$ and write $\hat{X}_\tau = P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$. Then for each $k \in \{1, \dots, \tau\}$,

$$\rho_k = \sum_{j=1}^{\tau} \alpha_{\tau,j} \rho_{k-j}.$$

Proof idea: decompose $\gamma_k = \text{Cov}(X_{\tau-k}, X_\tau) = \text{Cov}(X_{\tau-k}, \hat{X}_\tau) + \text{Cov}(X_{\tau-k}, X_\tau - \hat{X}_\tau)$; the second term is 0 by the orthogonality of the prediction error, and expanding the first using $\hat{X}_\tau = \sum_j \alpha_{\tau,j} X_{\tau-j}$ gives $\gamma_k = \sum_j \alpha_{\tau,j} \gamma_{k-j}$; divide by γ_0 .

Why: these are the **Yule–Walker-type** equations linking the ACF to the prediction coefficients $\alpha_{\tau,j}$. They are exactly what the Durbin–Levinson recursion solves to extract the PACF $\alpha_\tau = \alpha_{\tau,\tau}$.

Theorem 12: ACF & PACF signatures of ARMA models

For *causal and invertible* ARMA(p, q) models the ACF and PACF behave as:

	AR(p)	MA(q)	ARMA(p, q)
ACF	Tails off	Cuts off after lag q	Tails off
PACF	Cuts off after lag p	Tails off	Tails off

Proof idea: the ACF cut-off for MA(q) is the moment computation of Definition 3; the PACF

cut-off for $\text{AR}(p)$ is Proposition 13; “tails off” means a geometric/damped-sine decay with no exact zero (as for the $\text{AR}(1)$ ACF $\phi^{|\tau|}$).

Why: this 2×3 table is *the* model-identification tool: read q from where the ACF cuts, p from where the PACF cuts; if neither cuts (both tail off) the model is mixed ARMA.

Proposition 13: PACF of an $\text{AR}(p)$ vanishes beyond lag p

For a causal $\text{AR}(p)$,

$$\alpha_\tau = 0 \quad \text{for all } \tau > p.$$

Moreover the *estimated* PACF at lags $> p$ is, asymptotically, mean 0 with variance $1/n$ (n the sample size), giving the $\pm 1.96/\sqrt{n}$ significance bands.

Proof idea: for $\tau > p$ write the prediction error $X_\tau - \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$. Because the process is $\text{AR}(p)$, the true best predictor uses only $\phi_1, \dots, \phi_p$ and the noise ε_τ ; setting $\tilde{\phi}_j = \phi_j$ for $j \leq p$ and $\tilde{\phi}_j = 0$ for $j > p$, the prediction equations $\text{Cov}(X_\tau - \sum_j \alpha_{\tau,j} X_{\tau-j}, X_k) = 0$ are solved by $\alpha_{\tau,j} = \tilde{\phi}_j$ (using $\text{Cov}(\varepsilon_\tau, X_k) = 0$ for $k < \tau$, by causality). In particular $\alpha_\tau = \alpha_{\tau,\tau} = \tilde{\phi}_\tau = 0$ for $\tau > p$.

Why: this is the theoretical backbone of the table: it is why a sharp cut-off in the (sample) PACF at lag p diagnoses an $\text{AR}(p)$, and the $1/n$ variance gives the threshold for “cut off”.

Proposition 14: Non-identifiability of $\text{MA}(1)$ from the ACVS

The $\text{MA}(1)$ processes

$$X_t = \varepsilon_t - \theta \varepsilon_{t-1} \quad (\text{Var } \varepsilon_t = \sigma_\varepsilon^2) \quad \text{and} \quad Y_t = \eta_t - \frac{1}{\theta} \eta_{t-1} \quad (\text{Var } \eta_t = \sigma_\varepsilon^2 \theta^2)$$

have *identical* autocovariance sequences, hence are indistinguishable from second-order structure alone.

Proof idea: both give $\gamma_0 = \sigma_\varepsilon^2(1 + \theta^2)$, $\gamma_{\pm 1} = -\sigma_\varepsilon^2\theta$, and 0 elsewhere (by direct computation).

Why: an $\text{MA}(1)$ and its “root-reciprocal” twin share an ACVS; choosing the *invertible* representation ($|\theta| < 1$, root outside the unit circle) pins down a unique model. This is the reason invertibility is imposed.

3 ARMA Revisited: Backshift Operator, Wold Decomposition, Causality & Invertibility

Having met AR, MA and ARMA models as explicit sums of white noise, we now repackage them with the **backshift operator** B , which turns the difference equations into a clean algebra of *polynomials* $\Phi(z)$ and $\Theta(z)$. The payoff is conceptual: questions about a process (“is it stationary? is it invertible? what is its $\text{MA}(\infty)$ representation?”) become questions about the *roots* of these polynomials and the analyticity of a transfer function $G(z)$. Along the way the **Wold decomposition theorem** tells us that essentially every stationary process is a one-sided moving average of white noise, which is exactly why the linear/ARMA world is so central. Throughout, $\{\varepsilon_t\}$ is mean-zero white noise with $\text{Var}(\varepsilon_t) = \sigma^2$, and we use the lecture-note conventions: ARMA written $\Phi(B)X_t = \Theta(B)\varepsilon_t$, stationarity from roots of Φ *outside* the unit circle, invertibility from roots of Θ *outside* the unit circle.

3.1 Definitions

Definition 1: Backshift operator B

The **backshift operator** maps a time series $\{X_t\}$ to the series $\{Y_t\} = B[\{X_t\}]$ shifted back one step in time,

$$Y_t = X_{t-1} \quad \text{for all } t \in \mathbb{Z}.$$

We write informally $BX_t = X_{t-1}$. Iterating, $B^k X_t = X_{t-k}$ for $k \geq 0$, with $B^0 = I$ the identity. The operator is linear, and because shifts commute, products of polynomials in B may be multiplied and reordered like ordinary polynomials.

Definition 2: ARMA polynomial notation

An ARMA(p, q) process satisfies

$$\sum_{j=0}^p \phi_j X_{t-j} = \sum_{k=0}^q \theta_k \varepsilon_{t-k}, \quad \phi_0 = -1, \quad \theta_0 = -1,$$

which, with the backshift operator, becomes the compact statement

$$\Phi(B)X_t = \Theta(B)\varepsilon_t, \quad \Phi(z) = \sum_{j=0}^p \phi_j z^j, \quad \Theta(z) = \sum_{k=0}^q \theta_k z^k.$$

In the common “signed” form one writes $X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k \varepsilon_{t-k}$; simple examples instead use $X_t = \varepsilon_t - \theta \varepsilon_{t-1}$ etc. Pure cases: AR(p) is $\Phi(B)X_t = \varepsilon_t$ and MA(q) is $X_t = \Theta(B)\varepsilon_t$.

Definition 3: Characteristic polynomials

For $\Phi(B)X_t = \Theta(B)\varepsilon_t$ we call

- $\Phi(z)$ the **characteristic polynomial of the autoregressive part**, and
- $\Theta(z)$ the **characteristic polynomial of the moving-average part**.

For a pure AR, $\Phi(B)X_t = \varepsilon_t$, “the characteristic polynomial” means $\Phi(z)$; for a pure MA, $X_t = \Theta(B)\varepsilon_t$, it means $\Theta(z)$. The *roots* of these polynomials govern stationarity and invertibility.

Definition 4: Two-sided linear process

A **(two-sided) linear process** is

$$X_t = \sum_{k=-\infty}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 = \sum_k g_k^2 < \infty,$$

with $\{g_k\} \subset \mathbb{R}$ and $\{\varepsilon_t\}$ mean-zero white noise. Its first two moments are, for all $t, \tau \in \mathbb{Z}$,

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = \sigma^2 \|g\|_2^2 < \infty, \quad \text{Cov}(X_t, X_{t+\tau}) = \sigma^2 \sum_{k=-\infty}^{\infty} g_k g_{k+\tau}.$$

The summability $\|g\|_2^2 < \infty$ is what makes these moments finite, so the process is (second-order) stationary.

Definition 5: General linear process (one-sided)

If in the two-sided process we set $g_{-1} = g_{-2} = \dots = 0$, we obtain a **general linear process**

$$X_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k}, \quad \|g\|_2^2 < \infty.$$

Now X_t depends only on the *past and present* of $\{\varepsilon_t\}$, making the process **causal** (in the linear sense). The same mean/autocovariance formulas apply, so it is stationary. Writing $G(z) = \sum_{k \geq 0} g_k z^k$ gives the operator form $X_t = G(B)\varepsilon_t$. By convention $g_0 = 1$.

Definition 6: Transfer function $G(z)$, zeros and poles

For a linear process $X_t = G(B)\varepsilon_t$, the function

$$G(z) = \sum_{k=0}^{\infty} g_k z^k$$

is the **transfer function**. When G is a rational ratio $G(z) = G_1(z)/G_2(z)$:

- the roots of the numerator $G_1(z)$ are the **zeros** of G ;
- the roots of the denominator $G_2(z)$ are the **poles** of G .

For an ARMA model $G(z) = \Theta(z)/\Phi(z)$: the zeros of Φ (roots of the AR polynomial) are the poles of G , and the zeros of Θ (roots of the MA polynomial) are the zeros of G . Whether the expansion of G involves only past/present (versus future) noise is decided by where these roots sit relative to the unit circle $|z| = 1$.

Definition 7: Innovations outlier

If a single realised value of ε_t affects X_t and all subsequent $X_{t+1}, X_{t+2}, \dots$, that value is called an **innovations outlier**. This is exactly the behaviour of a one-sided (MA(∞)) process: each innovation ε_t propagates forward through the weights $g_0, g_1, \dots$.

3.2 Theorems & Propositions

Theorem 8: Wold decomposition theorem

Any (zero-mean) second-order stationary process $\{X_t\}$ can be written as

$$X_t = U_t + V_t, \quad U_t \text{ and } V_t \text{ uncorrelated,}$$

where

- U_t has a **one-sided linear** (MA(∞)) representation

$$U_t = \sum_{k=0}^{\infty} g_k \varepsilon_{t-k}, \quad g_0 = 1, \quad \|g\|_2^2 < \infty,$$

with $\{\varepsilon_t\}$ mean-zero white noise uncorrelated with V , i.e. $\mathbb{E}[\varepsilon_s V_t] = 0$ for all s, t ; the sequences $\{g_k\}$ and $\{\varepsilon_t\}$ are *uniquely* determined;

- V_t is **singular** (deterministic): it can be predicted from its own past with zero error.

Proof idea: project X_t onto the closed linear span of its own past; the residual one-step innovations are the white noise ε_t , the g_k are the projection coefficients, and V_t is the part lying in the infinite-past intersection of these spans (the purely deterministic component).

Why: this is the structural justification for the whole linear-model programme: up to a deterministic piece, *every* stationary process is a one-sided moving average of white noise, and ARMA models are the finite, rational approximations to this MA(∞).

Proposition 9: Laurent expansion of $1/G_2$ and the past/present criterion

Let $G(z) = G_1(z)/G_2(z)$ and let $z_1, \dots, z_p$ be the zeros of G_2 , ordered so that $z_1, \dots, z_k$ lie *inside* the unit circle and $z_{k+1}, \dots, z_p$ lie *outside*. A partial-fraction plus geometric-series (Laurent) expansion, convergent on $|z| = 1$, gives after replacing $z \mapsto B$

$$\frac{1}{G_2(B)} \varepsilon_t = \underbrace{\sum_{j=1}^k A_j \sum_{l=0}^{\infty} z_j^l \varepsilon_{t+1+l}}_{\text{future}} - \underbrace{\sum_{j=k+1}^p A_j \sum_{l=0}^{\infty} z_j^{-(l+1)} \varepsilon_{t-l}}_{\text{past+present}}.$$

Hence if *all* roots of $G_2(z)$ are outside the unit circle ($k = 0$), only past and present values appear, and the one-sided general linear process exists.

Proof idea: expand $A_j/(z - z_j)$ as a geometric series, choosing the *inside* expansion $\sum_l (z_j/z)^l$ for $|z_j| < 1$ and the *outside* expansion for $|z_j| > 1$, so that each series converges on $|z| = 1$.

Why: this is the precise mechanism behind “roots outside the unit circle $\Rightarrow$ causal”. It also gives the equivalent analytic statement: $|G(z)| < \infty$ for $|z| \leq 1$, i.e. G is **analytic inside and on the unit circle**.

Proposition 10: Stationarity of the general linear process

A general linear process $X_t = \sum_{k \geq 0} g_k \varepsilon_{t-k}$ with $\|g\|_2^2 < \infty$ is second-order stationary, with

$$\mathbb{E}[X_t] = 0, \quad \gamma_\tau = \sigma^2 \sum_{k=0}^{\infty} g_k g_{k+|\tau|}.$$

Proof idea: the mean is zero by linearity; bilinearity of covariance together with $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$ collapses the double sum to the single sum above, which is finite by Cauchy–Schwarz and $\|g\|_2^2 < \infty$.

Why: this is the engine for computing the ACVS of *any* causal ARMA once its MA(∞) weights $\{g_k\} = \{\psi_k\}$ are known.

Proposition 11: Stationarity (causality) of AR(p)

For AR(p), $\Phi(B)X_t = \varepsilon_t$, the (unique) stationary causal solution is $X_t = \Phi^{-1}(B)\varepsilon_t = G(B)\varepsilon_t$ with $G(z) = \Phi^{-1}(z)$. This power expansion is analytic on $|z| \leq 1$ iff all roots of $\Phi(z)$ lie *outside* the unit circle. Therefore

$$\text{AR}(p) \text{ is stationary} \iff \text{all roots of } \Phi(z) \text{ satisfy } |z| > 1.$$

Proof idea: the poles of $G = 1/\Phi$ are the roots of Φ ; by the Laurent criterion (Prop. 9) the one-sided expansion exists exactly when those poles are outside $|z| = 1$.

Why: this is *the* root test for AR stationarity; it reduces an infinite-order question to checking the moduli of p roots.

Proposition 12: AR(p) is always invertible; MA(q) is always stationary

- For AR(p), $\Phi(B)X_t = \varepsilon_t$ already expresses the white noise as a *finite* polynomial in X , so AR(p) is **always invertible** (the AR polynomial Φ has finite order; there is no MA polynomial, $\Theta \equiv 1$).
- For MA(q), $X_t = \Theta(B)\varepsilon_t = G(B)\varepsilon_t$ with $G(z) = \Theta(z)$ a *finite* polynomial, so $|G(z)| < \infty$ everywhere; hence MA(q) is **always stationary**.

Proof idea: “finite polynomial $\Rightarrow$ bounded on the unit disk”; only the *inverse* expansion can fail to converge, never the finite direct one.

Why: it tells you which check is *free*: never test an AR for invertibility or an MA for stationarity.

Proposition 13: Invertibility via G^{-1} and the MA(q) root test

A linear process $X_t = G(B)\varepsilon_t$ is **invertible** if it can be rewritten as an AR(∞),

$$G^{-1}(B)X_t = \varepsilon_t, \quad G^{-1}(z) = \sum_{k=0}^{\infty} h_k z^k,$$

with G^{-1} analytic on $|z| \leq 1$, equivalently $|G^{-1}(z)| < \infty$ for $|z| \leq 1$, equivalently *all poles of G^{-1} are outside* the unit circle (equivalently, all *zeros* of G are inside). For MA(q), $G = \Theta$, so

$$\text{MA}(q) \text{ is invertible} \iff \text{all roots of } \Theta(z) \text{ satisfy } |z| > 1.$$

Proof idea: inverting $X_t = \Theta(B)\varepsilon_t$ needs the power series of $1/\Theta$ to converge on $|z| \leq 1$; by the Laurent criterion this requires the roots of Θ (the poles of $1/\Theta$) outside the unit circle.

Why: invertibility is what makes the innovations ε_t recoverable from the observed past $X_t, X_{t-1}, \dots$, which is essential for forecasting and for parameter identifiability.

Corollary 14: Stationarity & invertibility summary for ARMA

For $\Phi(B)X_t = \Theta(B)\varepsilon_t$:

AR(p): stationary $\iff$ roots of Φ outside $|z| = 1$; *always* invertible.

MA(q): *always* stationary; invertible $\iff$ roots of Θ outside $|z| = 1$.

ARMA(p, q): stationary $\iff$ roots of Φ outside; invertible $\iff$ roots of Θ outside.

Why: the single most-used reference fact of the chapter; combine the two independent root tests.

Proposition 15: Causal AR(p) has $\alpha_\tau = 0$ for $\tau > p$

If $\{X_t\}$ is a causal AR(p), its partial autocorrelation function satisfies $\alpha_\tau = 0$ for all $\tau > p$.

Proof idea: the order- τ best linear predictor is $P_{X_0, \dots, X_{\tau-1}}(X_\tau) = \sum_{j=1}^{\tau} \alpha_{\tau,j} X_{\tau-j}$ and $\alpha_\tau = \alpha_{\tau,\tau}$.

Since $X_\tau = \sum_{j=1}^p \phi_j X_{\tau-j} + \varepsilon_\tau$, choosing $\alpha_{\tau,j} = \tilde{\phi}_j$ (with $\tilde{\phi}_j = \phi_j$ for $j \leq p$, $\tilde{\phi}_j = 0$ for $j > p$) makes the prediction residual equal ε_τ , which by causality satisfies $\text{Cov}(\varepsilon_\tau, X_k) = 0$ for all $k < \tau$ (here $\text{Cov}(X_t, \varepsilon_s) = 0$ when $t < s$). Thus the orthogonality (prediction) equations are met, so these are the true coefficients; in particular $\alpha_\tau = \alpha_{\tau,\tau} = \tilde{\phi}_\tau = 0$ for $\tau > p$.

Why: the PACF “cuts off” after lag p for an AR(p) — the diagnostic mirror image of the ACF cutting off after lag q for an MA(q) (this proof is Sheet 2, Ex. 2.7).

4 Parametric Estimation: Yule–Walker and Least Squares

Having modelled $\{X_t\}$ as an ARMA process, we now *fit* it: given data $X_1, \dots, X_n$, estimate the autoregressive coefficients $\phi_1, \dots, \phi_p$, the moving-average coefficients $\theta_1, \dots, \theta_q$ and the innovation variance σ_ε^2 . This part develops the two workhorse estimators of the course. The **Yule–Walker** (method-of-moments) estimator turns the AR recursion into a linear system in the autocovariances and is solved by a single matrix inversion (accelerated by the Levinson–Durbin algorithm). The **least-squares** estimator minimises the sum of squared one-step prediction errors; for pure AR models it is an ordinary linear regression (forward, backward, or forward–backward), and for general ARMA models it becomes a nonlinear minimisation in which the unobserved innovations $\{\varepsilon_t\}$ are reconstructed recursively. Throughout, $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2)$ is Gaussian white noise and the AR part is assumed causal (roots of $\Phi(z)$ outside the unit circle), so that ε_t is uncorrelated with the past X_{t-k} , $k > 0$ — the single fact that makes the whole derivation work.

4.1 Definitions

Definition 1: Yule–Walker estimators (AR(p))

Let $\{X_t\}$ be a mean-zero causal AR(p) process $X_t = \sum_{k=1}^p \phi_k X_{t-k} + \varepsilon_t$, observed at $X_1, \dots, X_n$. Form the sample ACVS (mean known to be zero)

$$\hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} X_t X_{t+|\tau|},$$

collect $\hat{\gamma}_p = (\hat{\gamma}_1, \dots, \hat{\gamma}_p)^\top$ and the symmetric Toeplitz matrix

$$\hat{\Gamma}_p = \begin{pmatrix} \hat{\gamma}_0 & \hat{\gamma}_1 & \cdots & \hat{\gamma}_{p-1} \\ \hat{\gamma}_1 & \hat{\gamma}_0 & \cdots & \hat{\gamma}_{p-2} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\gamma}_{p-1} & \hat{\gamma}_{p-2} & \cdots & \hat{\gamma}_0 \end{pmatrix}.$$

The **Yule–Walker estimators** are

$$\hat{\phi}_p = \hat{\Gamma}_p^{-1} \hat{\gamma}_p, \quad \hat{\sigma}_\varepsilon^2 = \hat{\gamma}_0 - \sum_{j=1}^p \hat{\phi}_j \hat{\gamma}_j.$$

They are the empirical analogue of the population identities $\gamma = \Gamma \phi$ and $\gamma_0 = \sum_j \phi_j \gamma_j + \sigma_\varepsilon^2$.

Definition 2: Forward least-squares estimators (AR(p))

Write the AR equations for the observable times $t = p+1, \dots, n$ as the linear model $\mathbf{X}_F = F\phi + \varepsilon$, where

$$\mathbf{X}_F = \begin{pmatrix} X_{p+1} \\ \vdots \\ X_n \end{pmatrix}, \quad F = \begin{pmatrix} X_p & X_{p-1} & \cdots & X_1 \\ X_{p+1} & X_p & \cdots & X_2 \\ \vdots & \vdots & \ddots & \vdots \\ X_{n-1} & X_{n-2} & \cdots & X_{n-p} \end{pmatrix}, \quad \phi = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_p \end{pmatrix}.$$

The **forward least-squares estimator** minimises

$$SS_F(\phi) = \|\mathbf{X}_F - F\phi\|^2 = \sum_{t=p+1}^n \left(X_t - \sum_{k=1}^p \phi_k X_{t-k} \right)^2 = \sum_{t=p+1}^n \varepsilon_t^2,$$

and equals the ordinary least-squares solution

$$\hat{\phi}_F = (F^\top F)^{-1} F^\top \mathbf{X}_F, \quad \hat{\sigma}_F^2 = \frac{\|\mathbf{X}_F - F\hat{\phi}_F\|^2}{n - 2p}.$$

The first p observations $X_1, \dots, X_p$ are discarded as conditioning values.

Definition 3: Backward least-squares estimators (AR(p))

Exploiting time reversibility of a Gaussian AR process, regress the *first* $n - p$ values on their *future* lags: for $t = 1, \dots, n - p$, $X_t = \sum_{j=1}^p \phi_j X_{t+j} + \nu_t$. In matrix form $\mathbf{X}_B = B\phi + \nu$ with

$$\mathbf{X}_B = \begin{pmatrix} X_1 \\ \vdots \\ X_{n-p} \end{pmatrix}, \quad B = \begin{pmatrix} X_2 & X_3 & \cdots & X_{p+1} \\ X_3 & X_4 & \cdots & X_{p+2} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n-p+1} & X_{n-p+2} & \cdots & X_n \end{pmatrix}.$$

The **backward least-squares estimator** minimises $SS_B(\phi) = \|\mathbf{X}_B - B\phi\|^2 = \sum_{t=1}^{n-p} (X_t - \sum_{k=1}^p \phi_k X_{t+k})^2$ and is

$$\hat{\phi}_B = (B^\top B)^{-1} B^\top \mathbf{X}_B, \quad \hat{\sigma}_B^2 = \frac{\|\mathbf{X}_B - B\hat{\phi}_B\|^2}{n - 2p}.$$

The same ϕ -parameters appear because $Y_t = X_{-t}$ is an AR(p) with identical coefficients and an innovation $\tilde{\nu}_t \stackrel{d}{=} \varepsilon_t$.

Definition 4: Forward-backward least-squares estimator

The **forward-backward** estimator $\hat{\phi}_{FB}$ is the minimiser of the *combined* criterion

$$SS_F(\phi) + SS_B(\phi).$$

Equivalently it is the OLS solution of the stacked system obtained by piling $\mathbf{X}_F, \mathbf{X}_B$ and F, B on top of each other. Simulation studies show it typically outperforms both the forward and the backward estimator.

Definition 5: Least squares for general ARMA via residual reconstruction

For an ARMA(p, q) model $\Phi(B)X_t = \Theta(B)\varepsilon_t$ the innovations $\{\varepsilon_t\}$ are unobserved. The least-squares estimator chooses the parameters minimising $S(\phi, \theta) = \sum_t \hat{\varepsilon}_t^2$, where for each candidate

parameter value the residuals are **reconstructed recursively** from the model equation after fixing the q pre-sample innovations (and any pre-sample data) to zero, e.g. $\varepsilon_0 = \varepsilon_{-1} = \dots = 0$. The minimisation is nonlinear (no closed form) except in the pure-AR case, where it reduces to the linear forward least squares above.

4.2 Theorems & Propositions

Proposition 6: Yule–Walker equations

Let $\{X_t\}$ be a mean-zero causal AR(p) process with ACVS γ_τ . Then for every $k \geq 1$,

$$\gamma_k = \sum_{j=1}^p \phi_j \gamma_{k-j},$$

and in particular, taking $k = 1, \dots, p$ and using $\gamma_{-\tau} = \gamma_\tau$,

$$\gamma = \Gamma \phi, \quad \gamma = (\gamma_1, \dots, \gamma_p)^\top, \quad \Gamma = (\gamma_{i-j})_{i,j=1}^p.$$

Moreover $\gamma_0 = \sum_{j=1}^p \phi_j \gamma_j + \sigma_\varepsilon^2$.

Proof idea: multiply $X_t = \sum_j \phi_j X_{t-j} + \varepsilon_t$ by X_{t-k} and take expectations. Because the process is *causal*, X_{t-k} depends only on $\varepsilon_{t-k}, \varepsilon_{t-k-1}, \dots$, so for $k > 0$ we have $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$; with mean zero $\mathbb{E}[X_{t-j} X_{t-k}] = \gamma_{k-j}$, giving the recursion. For the variance, multiply by X_t instead: the cross term becomes $\mathbb{E}[\varepsilon_t X_t] = \mathbb{E}[\varepsilon_t (\sum_j \phi_j X_{t-j} + \varepsilon_t)] = \sigma_\varepsilon^2$ (all $\mathbb{E}[\varepsilon_t X_{t-j}] = 0$, $j \geq 1$).

Why: these $p + 1$ equations express the $p + 1$ unknowns $(\phi_1, \dots, \phi_p, \sigma_\varepsilon^2)$ purely in terms of $\gamma_0, \dots, \gamma_p$; replacing the population ACVS by its sample version yields the Yule–Walker estimators. It is the canonical method-of-moments fit for AR models.

Proposition 7: Minimising the prediction MSE gives Yule–Walker

Let $\{Y_t\}$ be mean-zero stationary with ACVS γ . The coefficients $\phi_1, \dots, \phi_k$ minimising the one-step mean-square prediction error

$$\mathbb{E} \left[\left(Y_t - \sum_{i=1}^k \phi_i Y_{t-i} \right)^2 \right]$$

satisfy the Yule–Walker equations $\gamma_j = \sum_{i=1}^k \phi_i \gamma_{j-i}$ for $j = 1, \dots, k$.

Proof idea: since $\{Y_t\}$ is mean-zero, the objective equals the variance $\gamma_0 - 2 \sum_i \phi_i \gamma_i + \sum_{i,j} \phi_i \phi_j \gamma_{i-j}$, a positive quadratic in ϕ . Setting $\partial/\partial \phi_i = 0$ gives $-2\gamma_i + 2 \sum_j \phi_j \gamma_{i-j} = 0$, i.e. the Yule–Walker equations.

Why: it shows Yule–Walker is not merely a moment trick but the *best linear predictor*: the AR coefficients are exactly the optimal linear-prediction weights, which is why YW and least squares agree asymptotically for AR models.

Proposition 8: Forward / backward least squares are the OLS estimators

With the design matrices of Definitions 2–3, the minimisers of SS_F and SS_B are

$$\hat{\phi}_F = (F^\top F)^{-1} F^\top \mathbf{X}_F, \quad \hat{\phi}_B = (B^\top B)^{-1} B^\top \mathbf{X}_B,$$

with residual-variance estimators $\hat{\sigma}^2 = \|\text{residuals}\|^2 / (n - 2p)$.

Proof idea: each is a standard linear-regression sum of squares $\|\mathbf{y} - A\phi\|^2$; the normal equations $A^\top A \phi = A^\top \mathbf{y}$ give the stated solution, and the mean-square residual divides by the degrees of freedom $(n - p) - p$.

Why: it places AR estimation inside the familiar OLS framework, so all regression machinery

(existence, uniqueness when $A^\top A$ is invertible, $\hat{\sigma}^2$) applies directly.

Proposition 9: Time reversibility of a Gaussian AR process

If $\{X_t\}$ is a stationary Gaussian AR(p) process with coefficients $\phi_1, \dots, \phi_p$ and innovation variance σ_ε^2 , then the time-reversed process $Y_t = X_{-t}$ is an AR(p) process with the *same* coefficients and innovation variance:

$$Y_t = \sum_{j=1}^p \phi_j Y_{t-j} + \tilde{\nu}_t, \quad \tilde{\nu}_t \stackrel{d}{=} \varepsilon_t,$$

equivalently $X_t = \sum_{j=1}^p \phi_j X_{t+j} + \nu_t$ with $\nu_t = \tilde{\nu}_{-t}$.

Proof idea: a stationary Gaussian process is determined by its ACVS, and the ACVS is symmetric, $\gamma_{-\tau} = \gamma_\tau$; hence the forward and reversed processes have identical second-order structure and (being Gaussian) identical distributions. The full argument is given later via frequency-domain methods (the spectral density is even in f).

Why: it justifies the backward least-squares regression — predicting the present from the future uses the very same parameters — and underpins the forward–backward estimator.

Proposition 10: Levinson–Durbin: solving Yule–Walker in $O(p^2)$

The Yule–Walker system $\hat{\Gamma}_p \hat{\phi}_p = \hat{\gamma}_p$ has a symmetric Toeplitz coefficient matrix. The **Levinson–Durbin algorithm** exploits this structure to solve it recursively in $k = 1, \dots, p$, costing $O(p^2)$ operations rather than the $O(p^3)$ of a naive matrix inversion. As a by-product it produces the partial autocorrelations $\hat{\alpha}_k = \hat{\phi}_{k,k}$ and the prediction-error variances at every order.

Proof idea: the order- k solution is obtained from the order- $(k-1)$ solution by one rank-one update governed by the reflection coefficient $\hat{\phi}_{k,k} = (\hat{\gamma}_k - \sum_{j=1}^{k-1} \hat{\phi}_{k-1,j} \hat{\gamma}_{k-j}) / v_{k-1}$, where v_{k-1} is the running error variance; the remaining coefficients update as $\hat{\phi}_{k,j} = \hat{\phi}_{k-1,j} - \hat{\phi}_{k,k} \hat{\phi}_{k-1,k-j}$.

Why: for high-order AR fits, spectral estimation, or repeatedly solving nested systems, the quadratic cost (and the free PACF) makes Levinson–Durbin the standard solver.

Proposition 11: Least-squares fits need not be stationary

The forward, backward and forward–backward least-squares estimators $\hat{\phi}_F, \hat{\phi}_B, \hat{\phi}_{FB}$ are *not* constrained to yield a stationary (causal) model: the implied polynomial $\hat{\Phi}(z)$ may have roots inside the unit circle.

Proof idea: the OLS objective places no constraint on the root moduli; it only minimises the residual sum of squares, which can be smaller for a non-causal parameter vector.

Why: non-stationary estimates are a problem for *prediction*, but acceptable for some uses such as *spectral estimation*, where they still produce a valid spectral density estimate. The Yule–Walker estimator, by contrast, always yields a stationary fit.

5 Trends, Seasonality and Differencing: SARIMA

Real signals rarely look stationary: they drift (a *trend*) and they repeat (a *season*). The working model of this chapter is $X_t = \mu_t + Y_t$ with μ_t a deterministic time-dependent mean and $\{Y_t\}$ stationary. There are two ways to peel off μ_t : *estimate and subtract* (regression on t and on seasonal dummies), or *difference it away* with the operator $\nabla = I - B$. Differencing of order d annihilates a polynomial trend of degree $d-1$; a *seasonal* difference $\nabla_{(s)} = I - B^s$ kills a period- s season. Layering autoregressive and moving-average structure onto *both* the ordinary lags and the seasonal lags gives the SARMA family, and folding the differencing in front gives the (S)ARIMA

model $\Phi(B)(1-B)^d X_t = \Theta(B)\varepsilon_t$ — the standard Box–Jenkins workhorse. Throughout, $\{\varepsilon_t\}$ is mean-zero white noise of variance σ_ε^2 , AR/seasonal-AR coefficients are $\phi_j, \phi_j^{(s)}$ and MA/seasonal-MA coefficients are $\theta_k, \theta_k^{(s)}$.

5.1 Definitions

Definition 1: Mean-trend (signal-plus-noise) model

A series with a slowly varying level is modelled as

$$X_t = \mu_t + Y_t,$$

where μ_t is a deterministic, time-dependent mean (the **trend**) and $\{Y_t\}$ is a zero-mean second-order stationary process. The canonical linear case is $\mu_t = a + bt$; more generally μ_t may be a degree- $(q-1)$ polynomial in t . When the level instead repeats, $\mu_t = s_t$ with $s_{t+s} = s_t$ is a **periodic** (seasonal) trend of period s ; the two can coexist, $X_t = a + bt + s_t + Y_t$.

Definition 2: Difference operator ∇

The (first-)difference operator is

$$\nabla = I - B, \quad \nabla X_t = X_t - X_{t-1},$$

where B is the backshift operator $BX_t = X_{t-1}$. It is *linear*. Its d -th power is, by the binomial theorem,

$$\nabla^d = (I - B)^d = \sum_{k=0}^d \binom{d}{k} (-1)^k B^k, \quad \nabla^d X_t = \sum_{k=0}^d \binom{d}{k} (-1)^k X_{t-k}.$$

For example $\nabla^2 X_t = X_t - 2X_{t-1} + X_{t-2}$.

Definition 3: Seasonal difference operator $\nabla_{(s)}$

The **seasonal difference** of period s is

$$\nabla_{(s)} = I - B^s, \quad \nabla_{(s)} X_t = X_t - X_{t-s}.$$

It removes a deterministic component of period s (since $\nabla_{(s)} s_t = s_t - s_{t-s} = 0$) and, applied to a linear trend, leaves a constant: $\nabla_{(s)}(bt) = bs$. The ordinary difference is the case $s = 1$, $\nabla = \nabla_{(1)}$. To kill a linear-plus-seasonal trend one composes them, $\nabla \nabla_{(s)} = (I - B)(I - B^s)$.

Definition 4: Seasonal moving average $\text{SMA}(Q)_s$

A **seasonal moving average** of order Q at period s is

$$X_t = \varepsilon_t - \sum_{j=1}^Q \theta_j^{(s)} \varepsilon_{t-js} = \Theta^{(s)}(B)\varepsilon_t, \quad \Theta^{(s)}(z) = 1 - \sum_{j=1}^Q \theta_j^{(s)} z^{sj}.$$

Only lags that are *multiples of s* appear. The basic seasonal MA(1) is $X_t = \varepsilon_t - \theta \varepsilon_{t-s}$.

Definition 5: Seasonal autoregression SAR(P) _{s}

A **seasonal autoregression** of order P at period s is

$$X_t = \varepsilon_t + \sum_{j=1}^P \phi_j^{(s)} X_{t-js}, \quad \Phi^{(s)}(z) = 1 - \sum_{j=1}^P \phi_j^{(s)} z^{sj}, \quad \Phi^{(s)}(B)X_t = \varepsilon_t.$$

Again only the seasonal lags $s, 2s, \dots, Ps$ enter.

Definition 6: SARMA(p, q) $\times$ (P, Q) _{s}

Combining an ordinary ARMA(p, q) with a seasonal ARMA(P, Q) _{s} gives the **multiplicative seasonal ARMA**

$$\Phi(B) \Phi^{(s)}(B) X_t = \Theta(B) \Theta^{(s)}(B) \varepsilon_t,$$

with the four polynomials $\Phi(z) = 1 - \sum_{j=1}^p \phi_j z^j$, $\Theta(z) = 1 - \sum_{k=1}^q \theta_k z^k$, $\Phi^{(s)}(z) = 1 - \sum_{j=1}^P \phi_j^{(s)} z^{sj}$, $\Theta^{(s)}(z) = 1 - \sum_{k=1}^Q \theta_k^{(s)} z^{sk}$. Special cases: SMA(Q) _{s} is SARMA($0, 0$) $\times$ ($0, Q$) _{s} and SAR(P) _{s} is SARMA($0, 0$) $\times$ ($P, 0$) _{s} . Example, a SARMA($0, 1$) $\times$ ($0, 1$)₁₂:

$$X_t = (1 - \theta B)(1 - \theta^{(12)} B^{12}) \varepsilon_t.$$

Definition 7: ARIMA(p, d, q)

$\{X_t\}$ is an **ARIMA**(p, d, q) (“integrated ARMA”) process if its d -th difference is a (causal, invertible) ARMA(p, q):

$$\Phi(B) (1 - B)^d X_t = \Theta(B) \varepsilon_t.$$

Equivalently, writing $Z_t = \nabla^d X_t$,

$$Z_t = \sum_{j=1}^p \phi_j Z_{t-j} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j}.$$

The “I” stands for *integrated*: X_t is recovered from Z_t by summing (the inverse of differencing) d times. ARMA is the case $d = 0$.

Definition 8: SARIMA(p, d, q) $\times$ (P, D, Q) _{s}

The full **seasonal ARIMA** folds both an ordinary difference of order d and a seasonal difference of order D in front of a SARMA:

$$\Phi(B) \Phi^{(s)}(B) (1 - B)^d (1 - B^s)^D X_t = \Theta(B) \Theta^{(s)}(B) \varepsilon_t.$$

Here d removes a polynomial trend, D removes a (possibly stochastic) seasonal pattern, and the four polynomials capture the remaining short-range and seasonal dependence.

Definition 9: Random walk and the IMA(1, 1)/ARI(1, 1) models

The **random walk** is the AR(1) at the boundary $\phi = 1$,

$$X_t = X_{t-1} + \varepsilon_t,$$

which is *not* stationary, but whose first difference $\nabla X_t = \varepsilon_t$ is white noise: it is an ARIMA(0, 1, 0). Two minimal one-step-integrated models:

- **IMA**(1, 1) = ARIMA(0, 1, 1): $Z_t = \nabla Y_t = \varepsilon_t - \theta \varepsilon_{t-1}$, i.e. $Y_t = Y_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$;
- **ARI**(1, 1) = ARIMA(1, 1, 0): $Z_t = \nabla Y_t = \phi Z_{t-1} + \varepsilon_t$ with $|\phi| < 1$, i.e. $Y_t - Y_{t-1} = \phi(Y_{t-1} - Y_{t-2}) + \varepsilon_t$.

Neither $\{Y_t\}$ is stationary.

5.2 Theorems & Propositions

Proposition 10: ∇ removes a linear trend; ∇Y stays stationary

Let $X_t = a + bt + Y_t$ with $\{Y_t\}$ stationary. Then

$$\nabla X_t = X_t - X_{t-1} = b + \nabla Y_t,$$

so the trend $a + bt$ is reduced to the constant b , and the random part $\nabla Y_t = Y_t - Y_{t-1}$ is again stationary. Differencing once more removes the constant too: $\nabla^2 X_t = \nabla^2 Y_t = Y_t - 2Y_{t-1} + Y_{t-2}$.

Proof idea: substitute $a + bt + Y_t$ and cancel: $a + bt - (a + b(t-1)) = b$. Linearity of ∇ gives ∇Y_t ; a finite linear combination of a stationary process is stationary, hence so is ∇Y_t .

Why: the formal justification for first-differencing a series that “looks linear”; it converts a non-stationary level into a stationary increment, up to a known constant that the mean estimate then absorbs.

Proposition 11: ∇^q annihilates a degree- $(q-1)$ polynomial trend

Let $X_t = \mu_t + Y_t$ with μ_t a polynomial of degree $q-1$ in t . Then $\nabla^q \mu_t \equiv 0$, so

$$\nabla^q X_t = \nabla^q Y_t = \sum_{k=0}^q \binom{q}{k} (-1)^k Y_{t-k}.$$

Proof idea: ∇ lowers the degree of a polynomial in t by exactly one (the leading t^m becomes $t^m - (t-1)^m = m t^{m-1} + \dots$); applying ∇ a total of q times to a degree- $(q-1)$ polynomial drives it to the zero sequence. Linearity then leaves $\nabla^q Y_t$, expanded by the binomial theorem, using the induction fact that $\nabla^k t^{k-1} = 0$.

Why: tells you exactly how many differences $d = q$ are needed to flatten a polynomial trend of a given degree, which is how one chooses d in ARIMA(p, d, q).

Theorem 12: Combined $\nabla \nabla_{(12)}$ yields a stationary process

Let $X_t = a + bt + s_t + Y_t$, where $\{Y_t\}$ is zero-mean stationary with ACVS γ_τ , a, b are constants and $s_{t+12} = s_t$ has period 12. Then

$$\nabla \nabla_{(12)} X_t = \nabla \nabla_{(12)} Y_t = Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}$$

is weakly stationary, with mean 0 and

$$\begin{aligned} \text{Cov}(\nabla \nabla_{(12)} X_t, \nabla \nabla_{(12)} X_{t-\tau}) &= \gamma_\tau - \gamma_{\tau+1} - \gamma_{\tau+12} + \gamma_{\tau+13} \\ &\quad - \gamma_{\tau-1} + \gamma_\tau + \gamma_{\tau+11} - \gamma_{\tau+12} \\ &\quad - \gamma_{\tau-12} + \gamma_{\tau-11} + \gamma_\tau - \gamma_{\tau+1} \\ &\quad + \gamma_{\tau-13} - \gamma_{\tau-12} - \gamma_{\tau-1} + \gamma_\tau. \end{aligned}$$

Proof idea: first $\nabla_{(12)} X_t = 12b + \nabla_{(12)} Y_t$ kills s_t and turns bt into the constant $12b$; then ∇ kills the constant, leaving the fixed linear combination $Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}$. Its mean is 0 and its covariance, expanded by bilinearity, is a function of τ *only* (the four-by-four table above).

Why: the theoretical backing for the standard “ ∇_{12} then ∇ ” pre-processing of monthly data (e.g. the Mauna Loa CO₂ series); it shows no t -dependence survives.

Proposition 13: Multiplicative-trend variant: $\nabla_{(12)}^2$

If instead $X_t = (a + bt)s_t + Y_t$ with $s_{t+12} = s_t$, then

$$\nabla_{(12)} X_t = 12b s_{t-12} + Y_t - Y_{t-12}, \quad \nabla_{(12)}^2 X_t = Y_t - 2Y_{t-12} + Y_{t-24},$$

which has zero mean and a covariance depending only on $|\tau|$, hence is stationary.

Proof idea: $\nabla_{(12)}$ once turns $(a + bt)s_t$ into $12b s_{t-12}$ (using $s_t = s_{t-12}$); a second $\nabla_{(12)}$ removes that periodic term as well, leaving a fixed combination of Y 's.

Why: warns that when trend and season *multiply* (amplitude grows with level), a *single seasonal* double-difference — not $\nabla \nabla_{(12)}$ — is what flattens the series.

Proposition 14: ACVS of the seasonal MA(1)

For $X_t = \varepsilon_t - \theta \varepsilon_{t-s}$ with white-noise variance σ_ε^2 ,

$$\gamma_\tau = \sigma_\varepsilon^2 \left[(1 + \theta^2) \delta_\tau - \theta(\delta_{\tau-s} + \delta_{\tau+s}) \right] = \begin{cases} (1 + \theta^2) \sigma_\varepsilon^2, & \tau = 0, \\ -\theta \sigma_\varepsilon^2, & \tau = \pm s, \\ 0, & \text{otherwise,} \end{cases}$$

so the ACF is $\rho_{\pm s} = -\theta/(1 + \theta^2)$ and zero at every other lag.

Proof idea: apply the white-noise filter rule $\gamma_\tau = \sigma_\varepsilon^2 \sum_j a_j a_{j+|\tau|}$ to the taps $a_0 = 1$, $a_s = -\theta$; only $\tau \in \{0, \pm s\}$ pair surviving noise indices.

Why: a single nonzero ACF spike at the seasonal lag s is the fingerprint that flags an $\text{SMA}(1)_s$ component in a correlogram.

Proposition 15: Random walk is non-stationary; its difference is white noise

For the random walk $X_t = \sum_{j=1}^t \varepsilon_j$ (with $X_0 = 0$),

$$\mathbb{E}[X_t] = 0, \quad \text{Var}(X_t) = t \sigma_\varepsilon^2, \quad \text{Cov}(X_t, X_{t+\tau}) = t \sigma_\varepsilon^2 \quad (\tau \geq 0),$$

all of which depend on t , so $\{X_t\}$ is **not** stationary; but $\nabla X_t = \varepsilon_t$ is white noise.

Proof idea: the variance of a sum of t uncorrelated unit pieces grows linearly in t ; the overlap of X_t and $X_{t+\tau}$ is the first t shared terms.

Why: the prototype unit-root process — exactly the “ $\phi = 1$ boundary” that ARIMA differencing is designed to handle.

Theorem 16: Second-order growth of IMA(1, 1); correlation $\rightarrow 1$

Let $Y_t = Y_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$ with $Y_0 = 0$. Then for $t \geq 1$

$$Y_t = \varepsilon_t + (1 - \theta) \sum_{j=1}^{t-1} \varepsilon_j - \theta \varepsilon_0,$$

hence

$$\text{Var}(Y_t) = \sigma_\varepsilon^2 [1 + (1 - \theta)^2(t - 1) + \theta^2],$$

and for $\tau > 0$

$$\text{Cov}(Y_t, Y_{t+\tau}) = \sigma_\varepsilon^2 [1 - \theta + (1 - \theta)^2(t - 1) + \theta^2].$$

As $\tau/t \rightarrow 0$ the correlation $\text{Corr}(Y_t, Y_{t+\tau}) \rightarrow 1$.

Proof idea: unroll the recursion from $Y_0 = 0$ to get the explicit linear combination of $\varepsilon_0, \dots, \varepsilon_t$; read off Var and Cov from the shared coefficients (the bulk $(1 - \theta)$ terms dominate). Both grow like $(1 - \theta)^2(t - 1)\sigma_\varepsilon^2$, so their ratio $\rightarrow 1$.

Why: shows quantitatively that an integrated series has *variance growing linearly in t* and *nearby values almost perfectly correlated* — the diagnostic signature of needing a difference.

Theorem 17: Second-order growth of $\text{ARI}(1, 1)$; correlation near unity

Let $Y_t - Y_{t-1} = \phi(Y_{t-1} - Y_{t-2}) + \varepsilon_t$ with $|\phi| < 1$ and $Y_0 = 0$. Then

$$Y_t = \sum_{j=1}^t \sum_{i=0}^{t-j} \phi^i \varepsilon_j = \sum_{j=1}^t \frac{1 - \phi^{t-j+1}}{1 - \phi} \varepsilon_j,$$

so that

$$\begin{aligned} \text{Var}(Y_t) &= \sigma_\varepsilon^2 \sum_{j=1}^t \left(\frac{1 - \phi^{t-j+1}}{1 - \phi} \right)^2, \\ \text{Cov}(Y_t, Y_{t+\tau}) &= \sigma_\varepsilon^2 \sum_{j=1}^t \frac{1 - \phi^{t-j+1}}{1 - \phi} \cdot \frac{1 - \phi^{t+\tau-j+1}}{1 - \phi} \quad (\tau > 0). \end{aligned}$$

Again the correlation between nearby times is close to unity.

Proof idea: the increments $Z_t = \nabla Y_t$ form a stationary $\text{AR}(1)$, $Z_t = \sum_{i \geq 0} \phi^i \varepsilon_{t-i}$; integrating ($Y_t = \sum_{m \leq t} Z_m$ from 0) and summing the inner geometric series gives the closed coefficients $(1 - \phi^{t-j+1})/(1 - \phi)$.

Why: confirms that even when the increments are a well-behaved stationary $\text{AR}(1)$, *integrating* them recreates the random-walk-like variance growth and near-unit correlation, so the level $\{Y_t\}$ must be differenced before fitting.

6 Fourier Theory, Spectral Density and the Periodogram

The frequency domain is the second face of a stationary process. Every second-order property carried by the autocovariance sequence (ACVS) $\{\gamma_\tau\}$ in the time domain is mirrored, via the Fourier transform, by the **spectral density function** (SDF) $\mathcal{S}(f)$ in the frequency domain. This part builds that bridge in three stages. First, pure Fourier theory: continuous-time, discrete-time and finite-data transforms, the convolution theorem, aliasing and the Nyquist frequency. Second, the spectral representation of a stationary process: $\mathcal{S}(f) = \sum_\tau \gamma_\tau e^{-2\pi i f \tau}$, Herglotz's theorem, the integrated spectrum and its Lebesgue decomposition, and the spectral representation theorem with its orthogonal-increment process $\{Z(f)\}$. Third, estimation: the periodogram $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$, why it is (asymptotically) unbiased but *inconsistent*, and how tapering cures bias while multitapering cures variance. Throughout we take the sampling interval $\Delta = 1$ once we reach the spectral chapters, and we always work with ordinary frequency f (not angular frequency $\omega = 2\pi f$).

6.1 Definitions

Definition 1: Continuous-time Fourier transform and its inverse

For $g : \mathbb{R} \rightarrow \mathbb{C}$ with $g \in L^1$, the **Fourier transform** G (upper case denotes the transform; some authors write $\hat{g}$) is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-2\pi i f t} dt, \quad f \in \mathbb{R}.$$

If in addition $G \in L^1$, the **inverse transform** recovers g for almost all t :

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df.$$

We write $g \longleftrightarrow G$ and call (g, G) a **Fourier pair**.

Definition 2: Properties of the continuous-time transform

For $g, h \in L^1$ with transforms G, H and scalar α , the basic pairs are

$$\begin{aligned} \overline{g(t)} &\longleftrightarrow \overline{G(-f)}, & g(\alpha t) \ (\alpha \neq 0) &\longleftrightarrow \frac{1}{|\alpha|} G(f/\alpha), \\ g(t + \tau) &\longleftrightarrow G(f) e^{2\pi i \tau f}, & \alpha g(t) + h(t) &\longleftrightarrow \alpha G(f) + H(f). \end{aligned}$$

(Conjugation reflects, scaling in time inversely scales frequency, a time shift multiplies by a phase, and the transform is linear.)

Definition 3: Continuous-time convolution

For $g, h \in L^1$, the **convolution** $u = g * h$ is

$$u(t) = \int_{-\infty}^{\infty} g(t - u) h(u) du, \quad t \in \mathbb{R}.$$

Convolution at a point is a weighted average of one function by the other, centred at that point.

Definition 4: Discrete-time Fourier transform and its inverse

For a sequence $\{g_t\}_{t \in \Delta\mathbb{Z}}$ in ℓ^1 , the **discrete-time Fourier transform** is

$$G(f) = \Delta \sum_{t \in \Delta\mathbb{Z}} g_t e^{-2\pi i t f}, \quad f \in \mathbb{R},$$

with inverse (a Fourier series with the roles of domains swapped)

$$g_t = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i t f} df, \quad t \in \Delta\mathbb{Z}.$$

A defining feature is **periodicity**: $G(f) = G(f + k/\Delta)$ for every $k \in \mathbb{Z}$, so G is determined by its values on one period of length $1/\Delta$.

Definition 5: Discrete-time and periodic convolution

For $g, h \in \ell^1$ the **discrete convolution** $u = g * h$ is

$$u_t = \Delta \sum_{u \in \Delta\mathbb{Z}} g_{t-u} h_u, \quad t \in \Delta\mathbb{Z},$$

(with $\Delta = 1$ the prefactor disappears). The dual object, the **periodic convolution** $U = G * H$ of two periodic spectra, is

$$U(f) = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f - f') H(f') df'.$$

Definition 6: Finite-data Fourier transform and the Dirichlet kernel

Given observations $g_0, \dots, g_{(n-1)\Delta}$ at times $T = \{0, \Delta, \dots, (n-1)\Delta\}$, the **finite Fourier transform** is

$$G_n(f) = \sum_{t \in T} g_t e^{-2\pi i t f}, \quad f \in \mathbb{R}.$$

It equals the full transform of g_t multiplied by the indicator $d_t = \frac{1}{\Delta} \mathbf{1}_T(t)$, whose own transform is the **Dirichlet kernel**

$$\mathcal{D}_T(f) = \sum_{t \in T} e^{-2\pi i f t} = \frac{\sin(n\pi f \Delta)}{\sin(\pi f \Delta)} e^{-\pi i(n-1)f\Delta}.$$

Thus $G_n = \mathcal{D}_T * G$: observing a finite window *blurs* the true transform.

Definition 7: Fourier frequencies

The FFT evaluates G_n at the **Fourier frequencies**

$$f_k = \frac{k}{\Delta n}, \quad k \in \mathbb{Z}, \quad -\frac{n}{2} \leq k \leq \frac{n-1}{2}.$$

With $\Delta = 1$ these are $f_k = k/n$. They are equally spaced over one period and are the frequencies at which distinct periodogram ordinates are (approximately) uncorrelated.

Definition 8: Spectral density function (discrete and continuous)

For a discrete-time stationary process $\{X_t\}_{t \in \mathbb{Z}}$ with ACVS $\{\gamma_\tau\} \in \ell^1$, the **spectral density function** is the discrete-time Fourier transform of the ACVS (with $\Delta = 1$):

$$\mathcal{S}(f) = \sum_{\tau \in \mathbb{Z}} \gamma_\tau e^{-2\pi i f \tau}, \quad f \in \mathbb{R}.$$

For a continuous-time process with $\gamma \in L^1$, $\mathcal{S}(f) = \int_{-\infty}^{\infty} \gamma(\tau) e^{-2\pi i f \tau} d\tau$. By Fourier inversion the ACVS is recovered as $\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$; in particular $\gamma_0 = \text{Var}(X_t) = \int_{-1/2}^{1/2} \mathcal{S}(f) df$, so the SDF distributes the total “power” $\text{Var}(X_t)$ across frequencies. A valid SDF is real, even ($\mathcal{S}(-f) = \mathcal{S}(f)$, since $\gamma_{-\tau} = \gamma_\tau$), non-negative, and periodic with period 1.

Definition 9: Harmonic (sinusoidal) process

Let $A > 0$ and $\Theta \sim \text{Unif}(-\pi, \pi)$ be independent. The **harmonic process** is

$$X_t = A \cos(\nu t + \Theta), \quad t \in \mathbb{Z},$$

with $\nu \in \mathbb{R}$ the oscillation frequency. Then $\mathbb{E}[X_t] = 0$ and

$$\text{Cov}(X_{t+\tau}, X_t) = \frac{1}{2} \mathbb{E}[A^2] \cos(\nu \tau),$$

so it is stationary but its ACVS does *not* decay; in particular $\{\gamma_\tau\} \notin \ell^1$ and no ordinary SDF exists. Its spectrum is a pair of lines (jumps) at $\pm\nu/(2\pi)$.

Definition 10: Integrated spectrum (spectral distribution function)

By Herglotz’s theorem, every ACVS admits the representation

$$\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i \tau f} dS^{(I)}(f),$$

where $S^{(I)}$, the **integrated spectrum**, is right-continuous, bounded, non-decreasing, with $S^{(I)}(-\frac{1}{2}) = 0$ and $S^{(I)}(\frac{1}{2}) = \sigma^2 = \gamma_0$. It behaves like a (scaled) cumulative distribution function over frequency. When $\{\gamma_\tau\} \in \ell^1$ there is a density $\mathcal{S}$ with $S^{(I)}(f) = \int_{-1/2}^f \mathcal{S}(\lambda) d\lambda$, i.e. $\mathcal{S} = dS^{(I)}/df$, recovering Definition 8.

Definition 11: Periodogram

For data $X_1, \dots, X_n$ ($T = \{1, \dots, n\}$, $\Delta = 1$), the **periodogram** is the plug-in estimator obtained by Fourier transforming the biased sample ACVS:

$$\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau \in \mathbb{Z}} \hat{\gamma}_\tau e^{-2\pi i f \tau}, \quad \hat{\gamma}_\tau = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X})$$

for $|\tau| \leq n-1$ (0 otherwise). Equivalently (Proposition 23) $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$ with $J(f) = \sqrt{1/n} \sum_{t \in T} (X_t - \bar{X}) e^{-2\pi i t f}$.

Definition 12: Data taper and tapered periodogram

A **data taper** is a sequence $\{h_t\}_{t \in T}$ with $\|h\|_2^2 = \sum_t h_t^2 = 1$. The **tapered Fourier transform** and **tapered periodogram** are

$$J_h(f) = \sum_{t \in T} h_t (X_t - \bar{X}) e^{-2\pi i t f}, \quad \hat{\mathcal{S}}_h^{(p)}(f) = |J_h(f)|^2.$$

Choosing $h_t = \sqrt{1/n}$ (the constant taper) recovers the ordinary periodogram. The normalisation $\|h\|_2^2 = 1$ keeps the estimator unbiased for white noise.

Definition 13: Multitaper spectral estimator

Given K **orthonormal** tapers $h_k = \{h_{t,k}\}_{t \in T}$ with $\sum_{t \in T} h_{t,k} h_{t,k'} = \delta_{k,k'}$, the **multitaper estimator** averages the individual tapered periodograms,

$$\hat{\mathcal{S}}^{(mt)}(f) = \frac{1}{K} \sum_{k=1}^K \hat{\mathcal{S}}_{h_k}^{(p)}(f).$$

Its expectation is a convolution with the **average kernel** $\bar{H}(f) = \frac{1}{K} \sum_{k=1}^K |H_k(f)|^2$, and its variance is $\approx \mathcal{S}(f)^2/K$.

6.2 Theorems & Propositions

Theorem 14: Fourier inversion (continuous and discrete time)

(Continuous time.) If $g, G \in L^1$ then the inverse transform recovers g for almost all t : $g(t) = \int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df$. (Discrete time.) If $\{g_t\} \in \ell^1$ then G is recovered on one period as a Fourier series with the domains swapped:

$$g_t = \int_{-1/(2\Delta)}^{1/(2\Delta)} G(f) e^{2\pi i t f} df, \quad t \in \Delta\mathbb{Z}.$$

Proof idea: substitute the forward transform into the right-hand side and use the Fourier orthogonality relation ($\int e^{2\pi i f(t-s)} df$ acts as a delta), the integrability hypotheses justifying the interchange of integration/summation.

Why: inversion is what makes (g, G) a genuine *Fourier pair* — no information is lost. It is the engine behind recovering the ACVS from the SDF ($\gamma_\tau = \int_{-1/2}^{1/2} \mathcal{S}(f) e^{2\pi i f \tau} df$) and behind every “time $\leftrightarrow$ frequency” duality used below.

Theorem 15: Convolution theorem (continuous time)

For $g, h \in L^1$, if $u = g * h$ then $U = G \cdot H$: the transform of a convolution is the product of the transforms. If also $G, H \in L^1$ and $v = g \cdot h$, then dually $V = G * H$.

Proof idea: substitute the convolution integral into the definition of U , swap the order of integration (Fubini, justified by $\iint |g(x-y)h(y)| dx dy = \|g\|_1 \|h\|_1 < \infty$), and recognise the shift factor $e^{-2\pi i s f}$ as producing $G(f)$.

Why: the central computational identity of Fourier analysis — “multiplication in one domain is convolution in the other”. It underlies filtering, the SDF of a filtered process, and the periodogram bias formula.

Theorem 16: Convolution theorem (discrete time)

For $g, h \in \ell^1$ both pairs hold:

$$h \cdot g \longleftrightarrow H * G, \quad h * g \longleftrightarrow H \cdot G.$$

Proof idea: for $u = h * g$, expand $U(f) = \sum_t u_t e^{-2\pi i t f}$, insert $e^{-2\pi i (t-s)f} e^{-2\pi i s f}$, factor the double sum into $(\sum_s h_s e^{-2\pi i s f})(\sum_x g_x e^{-2\pi i x f})$; interchange is legal because $g, h \in \ell^1$. For $v = h \cdot g$, replace one factor by its inverse transform and integrate.

Why: the discrete analogue used to derive the finite-data transform ($G_n = \mathcal{D}_T * G$) and the periodogram expectation (a frequency convolution with the Fejér kernel).

Theorem 17: Aliasing theorem and the Nyquist frequency

Let $g \in L^1$ be continuous with transform G , and let G denote the transform of the sampled sequence $\{g(t)\}_{t \in \Delta\mathbb{Z}}$. Under the summability hypotheses, for every f

$$G(f) = \sum_{k \in \mathbb{Z}} G(f + k/\Delta).$$

Sampling *folds* all the continuous-frequency content at $\{f + k/\Delta\}_k$ onto the single frequency f . For an SDF this reads $\mathcal{S}(f) = \sum_{k \in \mathbb{Z}} \mathcal{S}(f + k)$ (with $\Delta = 1$). The **Nyquist frequency** is $1/(2\Delta)$ (i.e. $\frac{1}{2}$ when $\Delta = 1$): content above it cannot be recovered.

Proof idea: write $g(t)$ at integer times by both inversion formulas; split $\int_{-\infty}^{\infty} G(f) e^{2\pi i f t} df$ into period-length blocks, shift each block by k/Δ (the phase $e^{2\pi i t k/\Delta} = 1$ for $t \in \Delta\mathbb{Z}$), swap sum and integral (summability), and match the two representations of $g(t)$.

Why: explains why we sample fast enough; if $\mathcal{S}$ is negligible beyond $1/(2\Delta)$ the discrete spectrum faithfully matches the continuous one, otherwise it is corrupted (poor agreement).

Proposition 18: FFT complexity

The finite Fourier transform at the n Fourier frequencies $\{k/(\Delta n)\}$ can be computed in $O(n \log n)$ time via the **Fast Fourier Transform**, versus $O(n^2)$ for the naive sum.

Proof idea: divide-and-conquer — a transform of length n splits into two of length $n/2$ (even/odd indices), recursing $\log_2 n$ levels with $O(n)$ work each.

Why: makes periodogram/spectral computation feasible at scale; one can also invert the periodogram by FFT to get the sample ACVS cheaply.

Theorem 19: Herglotz's theorem

A complex sequence $\{g_t\}_{t \in \mathbb{Z}}$ is **non-negative definite** if and only if there exists a right-continuous, bounded, non-decreasing function $G^{(I)}$ with $G^{(I)}(-\frac{1}{2}) = 0$ such that

$$g_t = \int_{-1/2}^{1/2} e^{2\pi i t f} dG^{(I)}(f), \quad t \in \mathbb{Z}.$$

Proof idea: ($\Leftarrow$) any such integral representation yields a non-negative-definite sequence because $G^{(I)}$ is non-decreasing. ($\Rightarrow$) construct $G^{(I)}$ as the (weak) limit of the non-negative Cesàro/Fejér averages of the partial sums $\sum_{|t| \leq N} g_t e^{-2\pi i t f}$, which form a tight family of measures.

Why: it is the existence theorem for the spectrum. Since every ACVS is non-negative definite (a foundational property of any stationary process), *every* stationary process has an integrated spectrum $S^{(I)}$ — even those, like harmonic processes, with no ordinary SDF.

Corollary 20: Integrated spectrum exists for every stationary process

Applying Herglotz's theorem to the non-negative-definite ACVS gives $\gamma_\tau = \int_{-1/2}^{1/2} e^{2\pi i \tau f} dS^{(I)}(f)$ with $S^{(I)}(-\frac{1}{2}) = 0$, $S^{(I)}(\frac{1}{2}) = \sigma^2$, and $S^{(I)}$ non-decreasing.

Why: guarantees a frequency-domain description always exists; the SDF is merely the special (absolutely continuous) case $dS^{(I)} = \mathcal{S} df$.

Theorem 21: Lebesgue decomposition of the integrated spectrum

Any integrated spectrum decomposes uniquely as

$$S^{(I)}(\cdot) = S_1^{(I)}(\cdot) + S_2^{(I)}(\cdot) + S_3^{(I)}(\cdot),$$

where each $S_j^{(I)}$ is non-negative, non-decreasing with $S_j^{(I)}(-\frac{1}{2}) = 0$, and

1. $S_1^{(I)}$ is **absolutely continuous**: $S_1^{(I)}(f) = \int_{-1/2}^f \mathcal{S}(\lambda) d\lambda$ (the ordinary SDF $\mathcal{S}$);
2. $S_2^{(I)}$ is a **step function** with jumps $\{p_\ell\}$ at frequencies $\{f_\ell\}$ of pure sinusoids (a *line spectrum*);
3. $S_3^{(I)}$ is a **continuous singular** function (pathological, of no practical use).

Proof idea: the Lebesgue decomposition theorem for monotone functions / measures, specialised to the finite interval $[-\frac{1}{2}, \frac{1}{2}]$.

Why: classifies the possible spectral behaviours: purely continuous ($S_1^{(I)}$ only), purely discrete/line ($S_2^{(I)}$ only), or mixed.

Theorem 22: Spectral representation theorem

Let $\{X_t\}$ be a real-valued discrete-time stationary process with mean μ . There exists an **orthogonal-increment process** $\{Z(f)\}$ on $[-\frac{1}{2}, \frac{1}{2}]$ with

$$X_t = \mu + \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f) \quad (\text{equality in mean square}),$$

where for $f \neq f'$,

$$\mathbb{E}[dZ(f)] = 0, \quad \text{Var}(dZ(f)) = dS^{(I)}(f), \quad \text{Cov}(dZ(f), dZ(f')) = 0.$$

Proof idea: build Z as the L^2 (mean-square) limit of finite Fourier sums of the X_t 's; orthogonality of increments mirrors the non-decrease of $S^{(I)}$, and the variance of an increment is exactly the mass $S^{(I)}$ places there.

Why: the deep structural result: *every* stationary process is a superposition of uncorrelated random sinusoids $e^{2\pi i f t} dZ(f)$, with “amount of variance” $dS^{(I)}(f)$ at each frequency. It makes precise the slogan “ $\mathcal{S}(f) df$ is the power in $[f, f + df]$ ”.

Proposition 23: Periodogram as squared modulus of J

With $T = \{1, \dots, n\}$ and $Y_t = X_t - \bar{X}$,

$$\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2, \quad J(f) = \sqrt{\frac{1}{n}} \sum_{t \in T} Y_t e^{-2\pi i f t}.$$

Proof idea: starting from $\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau} \frac{1}{n} \sum_t Y_t Y_{t+|\tau|} e^{-2\pi i \tau f}$, re-index the double lag sum as a free double sum over $s, t \in T$,

$$\hat{\mathcal{S}}^{(p)}(f) = \frac{1}{n} \sum_{s=1}^n \sum_{t=1}^n Y_t Y_s e^{-2\pi i s f} e^{2\pi i t f} = \left(\sqrt{\frac{1}{n}} \sum_t Y_t e^{-2\pi i t f} \right) \overline{\left(\sqrt{\frac{1}{n}} \sum_s Y_s e^{-2\pi i s f} \right)} = |J(f)|^2.$$

Why: lets the periodogram be computed by a single FFT in $O(n \log n)$, and makes manifest that $\hat{\mathcal{S}}^{(p)} \geq 0$.

Theorem 24: Expectation of the periodogram: the Fejér kernel

For a zero-mean process,

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \sum_{\tau \in \mathbb{Z}} w_{\tau} \gamma_{\tau} e^{-2\pi i \tau f}, \quad w_{\tau} = \left(1 - \frac{|\tau|}{n}\right) \mathbf{1}_{[-n, n]}(\tau),$$

equivalently the frequency convolution

$$\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') \mathcal{F}_n(f - f') df', \quad \mathcal{F}_n(f) = \begin{cases} \frac{1}{n} \frac{\sin^2(n\pi f)}{\sin^2(\pi f)}, & f \neq 0, \\ n, & f = 0, \end{cases}$$

the **Fejér kernel**.

Proof idea: $\mathbb{E}[\hat{\gamma}_{\tau}] = (1 - \frac{|\tau|}{n}) \gamma_{\tau}$ for $|\tau| < n$; transform term by term. The triangular weight $w_{\tau} = \frac{1}{n} (d * \tilde{d})_{\tau}$ is the autocorrelation of the rectangular window, whose transform is $|\mathcal{D}_T(f)|^2 / n = \mathcal{F}_n(f)$, so multiplication by w_{τ} in time becomes convolution with $\mathcal{F}_n$ in frequency.

Why: quantifies the **bias**. Because $\mathcal{F}_n$ has wide side lobes that decay slowly, $\mathbb{E}[\hat{\mathcal{S}}^{(p)}]$ is a *blurred* (leaked) version of $\mathcal{S}$; peaks are smeared and power leaks into neighbouring frequencies. As $n \rightarrow \infty$, $\mathcal{F}_n$ concentrates at 0, so the periodogram is asymptotically unbiased.

Theorem 25: Variance of the periodogram: inconsistency

For $f \neq 0 \bmod \frac{1}{2}$,

$$\text{Var}(\hat{\mathcal{S}}^{(p)}(f)) \xrightarrow{n \rightarrow \infty} \mathcal{S}(f)^2 > 0,$$

and at distinct Fourier frequencies the ordinates are asymptotically uncorrelated.

Proof idea: for a Gaussian process $J(f)$ is asymptotically complex Gaussian, so $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$ is asymptotically $\frac{1}{2} \mathcal{S}(f) \chi_2^2 = \mathcal{S}(f) \cdot \text{Exp}(1)$, whose variance is $\mathcal{S}(f)^2$.

Why: the periodogram is **not consistent** — its variance does *not* vanish as n grows; more data merely gives more (equally noisy) ordinates. This is the motivation for smoothing/averaging (multitapering, or averaging neighbouring ordinates).

Theorem 26: Expectation of the tapered periodogram

With a taper $\{h_t\}$ (set $h_t = 0$ off T) and mean replaced by μ ,

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df',$$

where H is the discrete Fourier transform of h .

Proof idea: expand $\hat{\mathcal{S}}_h^{(p)}(f) = \sum_{t,s} h_t h_s Y_t Y_s e^{-2\pi i f(t-s)}$, take expectations to get $\sum_{\tau} w_{\tau} \gamma_{\tau} e^{-2\pi i f \tau}$ with $w_{\tau} = \sum_t h_t h_{t+\tau} = (\tilde{h} * \bar{h})_{\tau}$ (autocorrelation of the taper). By the convolution theorem its transform is $\tilde{H}(f) \bar{H}(f) = |H(f)|^2$, giving the frequency convolution.

Why: the kernel is now $|H(f)|^2$ instead of the Fejér kernel $\mathcal{F}_n$. A well-chosen taper (e.g. a dpss / Slepian taper) makes $|H(f)|^2$ far more concentrated near 0 with much lower side lobes, so **tapering reduces bias / leakage**.

Corollary 27: Unbiasedness of the tapered periodogram for white noise

If $\{X_t\}$ is white noise with variance σ^2 (so $\mathcal{S} \equiv \sigma^2$) and $\|h\|_2^2 = 1$, then $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 = \mathcal{S}(f)$ for all n and all f .

Proof idea: the kernel integrates out, $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2 \int_{-1/2}^{1/2} |H(f')|^2 df' = \sigma^2 \sum_t h_t^2 = \sigma^2 \|h\|_2^2$ (Parseval), which equals σ^2 exactly when $\|h\|_2^2 = 1$.

Why: pins down the normalisation $\|h\|_2^2 = 1$ in the definition of a taper, and shows tapering costs nothing in bias for flat spectra.

Theorem 28: Variance reduction by multitapering

For K orthonormal tapers, asymptotically

$$\text{Cov}(\hat{\mathcal{S}}_{h_k}^{(p)}(f), \hat{\mathcal{S}}_{h_{k'}}^{(p)}(f)) \rightarrow \mathcal{S}(f)^2 \delta_{k,k'}, \quad \text{hence} \quad \text{Var}(\hat{\mathcal{S}}^{(mt)}(f)) \approx \frac{\mathcal{S}(f)^2}{K}.$$

Proof idea: orthogonality of the tapers makes the K individual tapered periodograms asymptotically uncorrelated, each with variance $\mathcal{S}(f)^2$; averaging K uncorrelated terms divides the variance by K .

Why: multitapering restores consistency: as $n \rightarrow \infty$ one may let $K \rightarrow \infty$ (slowly) so $\text{Var} \rightarrow 0$. It is the standard fix for the periodogram's non-vanishing variance, with only a modest bias increase.

7 Multivariate Time Series and Vector Autoregression (VAR)

So far each X_t has been a scalar. Many real systems are d -dimensional: a basket of normalised stock prices (Apple, Nasdaq-100, S&P 500), or the longitude/latitude/temperature of an ocean drifter. We now let $\{X_t\}$ take values in $\mathbb{R}^d$ and ask two questions: (i) how do we *describe* the second-order structure — now a matrix-valued autocovariance and a matrix-valued spectrum, with genuine **cross**-terms between components, and (ii) how do we *model* it. The canonical model is the vector autoregression $\text{VAR}(p)$, and the key trick is that *every* $\text{VAR}(p)$ is a $\text{VAR}(1)$ in disguise (companion form), so its theory reduces to that of a single matrix Φ_1 (or F).

7.1 Definitions

Definition 1: Multivariate time series

A **multivariate time series** is a real d -vector-valued discrete-time stochastic process

$$X_t = (X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(d)})^\top, \quad t \in \mathbb{Z},$$

whose marginal processes $\{X_t^{(1)}\}, \dots, \{X_t^{(d)}\}$ are each univariate time series. When $d = 2$ we call it **bivariate**.

Definition 2: Multivariate second-order stationarity

$\{X_t\}$ is **multivariate second-order (weakly) stationary** if for all $t, s, \tau \in \mathbb{Z}$,

$$\mathbb{E}[X_t] = \mathbb{E}[X_s], \quad \text{Cov}(X_{t+\tau}, X_t) = \text{Cov}(X_{s+\tau}, X_s), \quad \text{tr}(\text{Var}(X_t)) < \infty.$$

Equivalently: each marginal $\{X_t^{(j)}\}$ is second-order stationary **and** every pair has a shift-invariant cross-covariance, $\text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)}) = \text{Cov}(X_{s+\tau}^{(j)}, X_s^{(k)})$ for all $1 \leq j, k \leq d$. (Here $\text{Cov}(U, V)$ of two vectors is the matrix with (j, k) entry $\text{Cov}(U^{(j)}, V^{(k)})$.)

Definition 3: Matrix autocovariance sequence (ACVS)

For a multivariate second-order stationary $\{X_t\}$, the **(matrix) autocovariance sequence** is the $d \times d$ matrix-valued sequence

$$\Gamma_\tau = \text{Cov}(X_{t+\tau}, X_t), \quad \tau \in \mathbb{Z},$$

with (j, k) entry $\gamma_\tau^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)})$.

- **The lag τ sits in the FIRST argument.** In the scalar case this is immaterial, but here it determines the convention for $\Gamma_{-\tau} = \Gamma_\tau^\top$ below.
- When $j = k$, $\gamma_\tau^{(j,j)}$ is the ordinary ACVS of the marginal $\{X_t^{(j)}\}$.
- When $j \neq k$, $\gamma_\tau^{(j,k)}$ is called the **cross-covariance sequence** (CCS) between components j and k .

Definition 4: Cross-correlation sequence (CCS)

The **cross-correlation sequence** between components j and k is

$$\rho_\tau^{(j,k)} = \frac{\gamma_\tau^{(j,k)}}{\sqrt{\gamma_0^{(j,j)} \gamma_0^{(k,k)}}}.$$

For $j = k$ this is the ordinary ACF (so $\rho_0^{(j,j)} = 1$). For $j \neq k$ it need *not* equal 1 at $\tau = 0$ and need *not* be symmetric in τ ; it satisfies $\rho_\tau^{(j,k)} = \rho_{-\tau}^{(k,j)}$.

Definition 5: Multivariate white noise $\text{WN}(0, \Sigma)$

A d -variate series $\{\varepsilon_t\}$ is **white noise with mean 0 and covariance matrix Σ** , written $\{\varepsilon_t\} \sim \text{WN}(0, \Sigma)$, iff it is stationary with mean vector 0 and

$$\Gamma_\tau^{(\varepsilon)} = \begin{cases} \Sigma & \tau = 0, \\ 0 & \tau \neq 0. \end{cases}$$

- There *can* be correlation between different components at lag 0 (off-diagonal entries of Σ):

the components are uncorrelated **across time** but possibly correlated **contemporaneously**.

- Often one further assumes Σ *diagonal* (componentwise independent noise), but this is an extra assumption, not part of the definition.

Definition 6: Uncorrelated processes

Two jointly second-order stationary processes $\{X_t^{(1)}\}$ and $\{X_t^{(2)}\}$ are **uncorrelated** if their cross-covariance vanishes at *every* lag:

$$\gamma_\tau^{(1,2)} = 0 \quad \text{for all } \tau \in \mathbb{Z}.$$

Definition 7: Spectral density matrix function

Assume $\{\gamma_\tau^{(j,k)}\}_{\tau \in \mathbb{Z}} \in \ell^1$ for all $1 \leq j, k \leq d$ (absolutely summable cross-covariances). The **spectral density matrix** is the $d \times d$ matrix-valued function

$$S(f) = \sum_{\tau \in \mathbb{Z}} \Gamma_\tau e^{-2\pi i \tau f}, \quad f \in [-\frac{1}{2}, \frac{1}{2}],$$

with inverse $\Gamma_\tau = \int_{-1/2}^{1/2} S(f) e^{2\pi i f \tau} df$.

- The (j, k) entry is $S_{j,k}(f)$. For $j = k$ it is the ordinary (real-valued) spectral density $S_j(f)$.
- For $j \neq k$, $S_{j,k}(f)$ is the **cross-spectral density**, which is *complex-valued* in general.

Definition 8: Coherence

The **coherence** between components j and k is

$$r_{j,k}(f) = \frac{S_{j,k}(f)}{\sqrt{S_{j,j}(f) S_{k,k}(f)}} \in \mathbb{C},$$

the frequency-domain analogue of a correlation between the increments dZ_j and dZ_k . One reports:

- its **magnitude** $|r_{j,k}(f)|$ (or magnitude-squared $|r_{j,k}(f)|^2 \in [0, 1]$), the strength of frequency-by-frequency association;
- its **phase** $\arg r_{j,k}(f)$, the “sign” / lead-lag angle in the complex plane.

Definition 9: Vector autoregression VAR(p)

Let $\{\varepsilon_t\} \sim \text{WN}(0, \Sigma)$ be d -variate. A process $\{X_t\}$ is a **vector autoregressive process of order p** , $\text{VAR}(p)$, if

$$X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \cdots + \Phi_p X_{t-p} + \varepsilon_t, \quad \Phi_j \in \mathbb{R}^{d \times d}, \Phi_p \neq 0.$$

Defining the matrix polynomial $\Phi(z) = I_d - \sum_{j=1}^p \Phi_j z^j$, this is written compactly as

$$\Phi(B)X_t = \varepsilon_t,$$

where B is the backshift operator. The same regressors (past values of the *whole* vector X_t) appear in every component equation; the off-diagonal entries of the Φ_j are exactly the cross-component dynamics.

Definition 10: Companion form of a VAR(p)

Stack the last p vectors into $Y_t = (X_t^\top, X_{t-1}^\top, \dots, X_{t-p+1}^\top)^\top \in \mathbb{R}^{dp}$. Then a VAR(p) is equivalently the VAR(1)

$$\underbrace{Y_t}_{dp \times 1} = \underbrace{F}_{dp \times dp} \underbrace{Y_{t-1}}_{dp \times 1} + \underbrace{U_t}_{dp \times 1}, \quad F = \begin{pmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I_d & 0 & \cdots & 0 & 0 \\ 0 & I_d & & \vdots & \vdots \\ \vdots & & \ddots & 0 & 0 \\ 0 & \cdots & 0 & I_d & 0 \end{pmatrix}, \quad U_t = \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The top block row reproduces the VAR(p) equation; the lower rows are identities $X_{t-i} = X_{t-i}$ that shift the stack. Recover the original process via $X_t = GY_t$ with $G = (I_d \ 0 \ \cdots \ 0) \in \mathbb{R}^{d \times dp}$.

Definition 11: Stability of a VAR(p)

A VAR(p) is **stable** if the roots of the **reverse characteristic polynomial**

$$\det(I_d - \Phi_1 z - \Phi_2 z^2 - \cdots - \Phi_p z^p) = 0$$

all lie *strictly outside* the complex unit circle ($|z| > 1$). Equivalently, the companion matrix F has all eigenvalues of modulus *strictly less than one* ($|\lambda| < 1$). Stability $\Rightarrow$ stationarity (but not conversely in general).

7.2 Theorems & Propositions

Proposition 12: Symmetry of the matrix ACVS

For a multivariate second-order stationary process,

$$\gamma_\tau^{(j,k)} = \gamma_{-\tau}^{(k,j)} \quad \Longleftrightarrow \quad \Gamma_{-\tau} = \Gamma_\tau^\top.$$

Proof idea: $\gamma_\tau^{(j,k)} = \text{Cov}(X_{t+\tau}^{(j)}, X_t^{(k)}) = \text{Cov}(X_t^{(k)}, X_{t+\tau}^{(j)})$ (covariance is symmetric in its arguments) $= \text{Cov}(X_{s-\tau}^{(k)}, X_s^{(j)})$ (stationarity, set $s = t + \tau$) $= \gamma_{-\tau}^{(k,j)}$.

Why: the cross-covariance need *not* be even in τ ($\gamma_\tau^{(j,k)} \neq \gamma_{-\tau}^{(j,k)}$ in general); only the *transpose* symmetry $\Gamma_{-\tau} = \Gamma_\tau^\top$ holds. This asymmetry is precisely what encodes lead-lag relationships between channels.

Proposition 13: Positive semi-definiteness of the matrix ACVS

The matrix sequence $\{\Gamma_\tau\}$ is positive semi-definite: for every $n \in \mathbb{N}$ and every $a_1, \dots, a_n \in \mathbb{R}^d$,

$$\sum_{j=1}^n \sum_{k=1}^n a_j^\top \Gamma_{k-j} a_k \geq 0.$$

Proof idea: the left side equals $\text{Var}(\sum_j a_j^\top X_{t_j}) \geq 0$ for suitable times (the scalar random variable $\sum_j a_j^\top X_{t_j}$ has non-negative variance).

Why: this is the defining algebraic constraint a valid matrix ACVS must satisfy; a candidate matrix sequence violating it cannot be the autocovariance of any process.

Proposition 14: Contemporaneous correlation induced by shared noise

Let $X_t = (X_t^{(1)}, X_t^{(2)})^\top$ be second-order stationary with ACVS $\Gamma_\tau^{(X)}$, and let $\{\phi_t\}$ be a univariate mean-zero white noise (variance σ_ϕ^2) independent of $\{X_t\}$. Define $Y_t = X_t + \phi_t \mathbf{1}$ (the *same* scalar noise added to both components), where $\mathbf{1} = (1, 1)^\top$. Then

$$\Gamma_\tau^{(Y)} = \Gamma_\tau^{(X)} + \sigma_\phi^2 \delta_{\tau,0} \mathbf{1}\mathbf{1}^\top, \quad \mathbf{1}\mathbf{1}^\top = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

Proof idea: $\Gamma_\tau^{(Y)} = \text{Cov}(X_{t+\tau} + \phi_{t+\tau}\mathbf{1}, X_t + \phi_t\mathbf{1})$; independence kills the X - ϕ cross terms, and $\text{Cov}(\phi_{t+\tau}, \phi_t) = \sigma_\phi^2 \delta_{\tau,0}$ contributes $\sigma_\phi^2 \delta_{\tau,0}$ to *every* entry.

Why: adding a common shock creates lag-0 correlation between $Y^{(1)}$ and $Y^{(2)}$ *regardless* of the correlation in X . This is the prototypical mechanism behind “everything moves together” contemporaneous correlation.

Proposition 15: Hermitian symmetry of the spectral matrix

For all $f \in [-\frac{1}{2}, \frac{1}{2}]$ and all $1 \leq j, k \leq d$,

$$S_{j,k}(f) = \overline{S_{k,j}(f)}, \quad S_{j,k}(f) = \overline{S_{j,k}(-f)},$$

equivalently in matrix form

$$S(f) = S(f)^H, \quad S(f) = S(-f)^\top,$$

and $S(f)$ is positive semi-definite for every f . *Proof idea:* substitute $\Gamma_{-\tau} = \Gamma_\tau^\top$ into $S(f) = \sum_\tau \Gamma_\tau e^{-2\pi i \tau f}$ and relabel $\tau \rightarrow -\tau$; the conjugate of $e^{-2\pi i \tau f}$ flips the sign in the exponent.

Why: $S(f)^H = S(f)$ (Hermitian) guarantees real diagonal spectra and makes coherence well defined; positive semi-definiteness is the frequency-domain counterpart of Proposition 13.

Theorem 16: Multivariate spectral representation theorem

Let $\{X_t\}$ be a vector-valued discrete-time stationary process with mean μ . There exists a vector-valued orthogonal-increment process $\{Z(f)\}$ on $[-\frac{1}{2}, \frac{1}{2}]$ such that

$$X_t = \mu + \int_{-1/2}^{1/2} e^{2\pi i f t} dZ(f).$$

If moreover $\{\gamma_\tau^{(j,k)}\} \in \ell^1$, then for $f, f' \in [-\frac{1}{2}, \frac{1}{2}]$,

$$\mathbb{E}[dZ(f)] = 0, \quad \text{Var}(dZ(f)) = S(f) df, \quad \text{Cov}(dZ(f), dZ(f')) = 0 \quad (f \neq f'),$$

where $S(f)$ is the spectral density *matrix* and the 0 in the last identity is the zero matrix.

Proof idea: the multivariate analogue of the scalar Cramér/spectral representation; the increment process $Z(f)$ carries the random “Fourier content” of each component, with covariance structure given by the spectral matrix.

Why: it justifies thinking of a multivariate stationary process as a superposition of uncorrelated random sinusoids whose (matrix) variance is $S(f) df$; coherence is then literally the correlation of the increments $dZ_j(f), dZ_k(f)$.

Proposition 17: Cross-spectrum of a pure delay process

Let $\{Y_t\}$ be stationary and $X_t = (Y_t, Y_{t-\nu})^\top$ for fixed $\nu \in \mathbb{Z}$. Then

$$S_{1,1}(f) = S_{2,2}(f) = \mathcal{S}_Y(f), \quad S_{1,2}(f) = \mathcal{S}_Y(f) e^{2\pi i f \nu},$$

so the coherence is $r_{1,2}(f) = e^{2\pi i f \nu}$, of magnitude 1 and phase $2\pi \nu f$. *Proof idea:* $\gamma_\tau^{(1,2)} = \text{Cov}(Y_{t+\tau}, Y_{t-\nu}) = \gamma_{\tau+\nu}^{(Y)}$; Fourier-transforming and shifting the summation index pulls out $e^{2\pi i f \nu}$.

Why: a pure time shift between two channels produces *perfect* coherence (magnitude 1) with a phase that is *linear in f* with slope $2\pi \nu$ — the textbook signature of a lead/lag of ν samples.

Theorem 18: Everything is a VAR(1) (companion reduction)

Any VAR(p) process $\{X_t\}$ admits the VAR(1) representation $Y_t = FY_{t-1} + U_t$ of Definition 10, with $\{U_t\}$ a dp -variate white noise whose only nonzero covariance block is the top-left Σ . Hence all VAR(p) properties (stability, MA(∞) expansion, covariances) follow from the VAR(1) theory applied to F and then projected back by $X_t = GY_t$.

Proof idea: write out the stacked vector Y_t ; its top block equals the VAR(p) recursion and its remaining blocks are trivial identities, giving exactly F and U_t .

Why: it lets us prove one clean VAR(1) result and reuse it for every order p , e.g. the stability test becomes “eigenvalues of F inside the unit circle”.

Proposition 19: MA(∞) representation of a stable VAR(1)

If the VAR(1) $X_t = \Phi_1 X_{t-1} + \varepsilon_t$ is stable, then

$$X_t = \sum_{j=0}^{\infty} \Phi_1^j \varepsilon_{t-j}.$$

Proof idea: back-substitute repeatedly, $X_t = \Phi_1(\Phi_1 X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \Phi_1^2 X_{t-2} + \Phi_1 \varepsilon_{t-1} + \varepsilon_t = \dots$; stability ($\Phi_1^j \rightarrow 0$) makes the remainder term vanish and the series converge (in mean square).

Why: the matrix geometric series is the multivariate causal linear filter of the noise — it underlies the covariance formula below and all forecasting.

Proposition 20: Covariance of a stable VAR(1)

For a stable VAR(1) with $\{\varepsilon_t\} \sim \text{WN}(0, \Sigma)$ and $\tau \geq 0$,

$$\Gamma_\tau = \text{Cov}(X_{t+\tau}, X_t) = \sum_{k=0}^{\infty} \Phi_1^{k+\tau} \Sigma (\Phi_1^k)^\top.$$

Proof idea: substitute the MA(∞) form into $\text{Cov}(\sum_j \Phi_1^j \varepsilon_{t+\tau-j}, \sum_k \Phi_1^k \varepsilon_{t-k})$; white noise makes only the terms with $t + \tau - j = t - k$, i.e. $j = k + \tau$, survive, each contributing $\Phi_1^{k+\tau} \Sigma (\Phi_1^k)^\top$.

Why: closed-form second-order structure of a VAR(1); for $\tau = 0$ it gives $\Gamma_0 = \sum_k \Phi_1^k \Sigma (\Phi_1^k)^\top$, the stationary covariance, which also solves the discrete Lyapunov equation $\Gamma_0 = \Phi_1 \Gamma_0 \Phi_1^\top + \Sigma$.

Corollary 21: Covariance of a VAR(p) via the companion form

For a stable VAR(p) with companion matrix F and $\Sigma_U = \text{Var}(U_t)$ (top-left block Σ , all else 0),

$$\Gamma_\tau^{(X)} = \text{Cov}(X_{t+\tau}, X_t) = G \left(\sum_{k=0}^{\infty} F^{k+\tau} \Sigma_U (F^k)^\top \right) G^\top, \quad G = (I_d \ 0 \ \dots \ 0).$$

Proof idea: apply Proposition 20 to the VAR(1) $Y_t = FY_{t-1} + U_t$ to get $\Gamma_\tau^{(Y)}$, then use $X_t = GY_t$ so $\Gamma_\tau^{(X)} = G \Gamma_\tau^{(Y)} G^\top$.

Why: reduces the order- p covariance to one VAR(1) computation plus a projection.

Theorem 22: Yule–Walker equations for a VAR(p)

For a stable VAR(p) and $\tau \geq 0$,

$$\Gamma_\tau = \Phi_1 \Gamma_{\tau-1} + \Phi_2 \Gamma_{\tau-2} + \cdots + \Phi_p \Gamma_{\tau-p} + \delta_{\tau,0} \Sigma.$$

In particular for VAR(1): $\Gamma_\tau = \Phi_1 \Gamma_{\tau-1} + \delta_{\tau,0} \Sigma$. *Proof idea:* multiply the VAR equation on the right by $X_t^\top$ and take expectations: $\text{Cov}(X_{t+\tau}, X_t) = \text{Cov}(\sum_i \Phi_i X_{t+\tau-i} + \varepsilon_{t+\tau}, X_t)$; the noise term $\text{Cov}(\varepsilon_{t+\tau}, X_t)$ contributes Σ only when $\tau = 0$ (since X_t depends only on ε_s for $s \leq t$). *Why:* given Σ and the Φ_j , solving the first $p+1$ equations yields $\Gamma_0, \dots, \Gamma_p$, and the recursion then generates all Γ_τ for $\tau > p$. Run in reverse, it estimates the Φ_j from sample covariances (VAR fitting).

8 Forecasting

The purpose of time series analysis is often to *forecast* future values from the data observed up to now. The recipe has three steps: (1) assume the form of the model; (2) estimate its parameters; (3) use the assumed structure and the estimated parameters to form predictions. In this part we take the model structure and the parameters as **known** and predict h lags ahead, i.e. we want X_{n+h} from $X_1, \dots, X_n$. The integer $h > 0$ is the **lead** (or **horizon**) and n is the **origin**. The central theme is that, for the second-order theory we use, the optimal predictor depends *only* on the covariance structure $\{\gamma_\tau\}$, and for causal ARMA models it is computed cleanly by setting future innovations to zero.

8.1 Definitions

Definition 1: Prediction mean square error

Let $g(X_1, \dots, X_n)$ be any estimator (predictor) of X_{n+h} based on the observations $X_1, \dots, X_n$. Its performance is measured by the **prediction mean square error**

$$P_{n+h}^n := \mathbb{E} \left[(X_{n+h} - g(X_1, \dots, X_n))^2 \right].$$

Smaller P_{n+h}^n means a better forecast; minimising it over a class of predictors defines the “best” predictor in that class.

Definition 2: Best linear predictor

The **best linear predictor** of X_{n+h} based on $X_1, \dots, X_n$ is the linear function

$$X_{n+h}^n = P_{X_1, \dots, X_n}(X_{n+h}) = \sum_{j=1}^n \beta_j X_j$$

that minimises the prediction mean square error among all linear combinations of $X_1, \dots, X_n$. We write $P_{X_1, \dots, X_n}(\cdot)$ for the projection onto the linear span of the data.

- For **Gaussian** time series the conditional expectation $\mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$ is itself linear in the data, so the best linear predictor coincides with the (overall) best predictor.
- In general the conditional expectation is a *nonlinear* function of the data; for tractability one restricts attention to linear predictors, which need only second-order information.

Definition 3: h -step forecast error and its variance

For the best linear predictor X_{n+h}^n define the **h -step ahead forecast error**

$$e_n(h) := X_{n+h} - X_{n+h}^n.$$

It is unbiased, $\mathbb{E}[e_n(h)] = 0$, and its variance equals the prediction MSE, $\text{Var}(e_n(h)) = P_{n+h}^n$. For a causal ARMA process with $MA(\infty)$ weights $\{\psi_j\}$ one has $e_n(h) = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j}$ and $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$.

Definition 4: Sources of prediction uncertainty

There are **three** components of uncertainty in a forecast:

1. **model uncertainty** — the assumed model form may be wrong;
2. **estimation uncertainty** — the parameters are estimated, not known;
3. **innovation uncertainty** — the inherent randomness of the future innovations $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$.

Throughout we assume the model is *known* and the parameters are estimated *without error*, which removes the first two sources and leaves only innovation uncertainty. (Bootstrap methods can incorporate the first two, but are not covered here.)

Definition 5: Gaussian prediction interval

For a **Gaussian** process the h -step error $e_n(h)$ is normal with mean 0 and variance $\text{Var}(e_n(h))$, so a $(1 - \alpha)$ **prediction interval** for X_{n+h} based on $X_1, \dots, X_n$ is

$$X_{n+h}^n \pm z_{1-\alpha/2} \sqrt{\text{Var}(e_n(h))},$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard normal distribution (e.g. $z_{0.975} \approx 1.96$ for a 95% interval).

8.2 Theorems & Propositions

Lemma 6: Conditional expectation minimises the prediction MSE

The conditional expectation

$$g(X_1, \dots, X_n) = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$$

minimises the prediction mean square error for X_{n+h} based on $X_1, \dots, X_n$.

Proof idea: For any real random variable Y with $\mu = \mathbb{E}[Y]$ and any constant c ,

$$\text{MSE}(c) = \mathbb{E}[(Y - c)^2] = \mathbb{E}[(Y - \mu + \mu - c)^2] = \mathbb{E}[(Y - \mu)^2] + (\mu - c)^2 = \text{Var}(Y) + (\mu - c)^2,$$

which is minimised at $c = \mu = \mathbb{E}[Y]$ (the cross term vanishes because $\mathbb{E}[Y - \mu] = 0$). Applying this *conditionally on* $X_1, \dots, X_n$ and using the tower property $\mathbb{E}[(X_{n+h} - g)^2] = \mathbb{E}[\mathbb{E}[(X_{n+h} - g)^2 \mid X_1, \dots, X_n]]$, the inner MSE is minimised pointwise by taking $g = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n]$.

Why: this is the theoretical gold standard for forecasting. For Gaussian processes the conditional expectation is linear, so the best predictor *is* the best linear predictor; in general it motivates restricting to linear predictors, which only need the ACVS.

Theorem 7: Prediction equations (zero-mean stationary process)

For a zero-mean stationary process $\{X_t\}$, the best linear predictor $X_{n+h}^n = \sum_{j=1}^n \beta_j X_j$ is found by solving for $\beta_1, \dots, \beta_n$ the **prediction equations**

$$\mathbb{E}[(X_{n+h} - X_{n+h}^n) X_k] = 0, \quad k = 1, \dots, n,$$

i.e. the forecast error is orthogonal to every observation. Equivalently, in matrix form $\Gamma_n \beta = \gamma_{[h]}$:

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{n-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{n-1} & \gamma_{n-2} & \cdots & \gamma_0 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix} = \begin{pmatrix} \gamma_{n+h-1} \\ \gamma_{n+h-2} \\ \vdots \\ \gamma_h \end{pmatrix}.$$

Proof idea: the best linear predictor is the orthogonal projection of X_{n+h} onto $\text{span}\{X_1, \dots, X_n\}$; the residual $X_{n+h} - X_{n+h}^n$ must be orthogonal (uncorrelated) to each X_k . Expanding $\mathbb{E}[(X_{n+h} - \sum_j \beta_j X_j) X_k] = 0$ gives $\gamma_{n+h-k} = \sum_{j=1}^n \beta_j \gamma_{j-k}$, which is row k of $\Gamma_n \beta = \gamma_{[h]}$.

Why: this is the master equation for linear prediction of *any* stationary process — it shows the optimal forecast uses only the ACVS.

Corollary 8: Matrix solution and prediction MSE

If Γ_n is invertible, then with $X = (X_1, \dots, X_n)^\top$ the best linear predictor and its MSE are

$$X_{n+h}^n = \gamma_{[h]}^\top \Gamma_n^{-1} X, \quad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

Proof idea: solve $\Gamma_n \beta = \gamma_{[h]}$ for $\beta = \Gamma_n^{-1} \gamma_{[h]}$ and substitute into $X_{n+h}^n = \beta^\top X$ and into $P_{n+h}^n = \gamma_0 - 2\beta^\top \gamma_{[h]} + \beta^\top \Gamma_n \beta$; the last two terms collapse using $\Gamma_n \beta = \gamma_{[h]}$.

Why: gives the closed-form forecast for short series.

- Γ_n is invertible for *invertible* ARMA models (and not in all other cases), but the best prediction X_{n+h}^n is **always unique** even when Γ_n is singular.
- Computing $\gamma_{[h]}^\top \Gamma_n^{-1} X$ is inefficient for large n because the $n \times n$ matrix must be inverted; the Durbin–Levinson and innovations algorithms avoid this (Shumway & Stoffer, §3.5).

Theorem 9: Prediction equations with non-zero mean (and intercept β_0)

For a stationary process with $\mu = \mathbb{E}[X_t]$, write $X_{n+h}^n = \beta_0 + \sum_{j=1}^n \beta_j X_j$. The coefficients solve the prediction equations $\mathbb{E}[(X_{n+h} - X_{n+h}^n) X_k] = 0$ for $k = 0, 1, \dots, n$ (with the convention $X_0 \equiv 1$), which give

$$\Gamma_n \beta = \gamma_{[h]}, \quad \beta_0 = \mu(1 - \beta^\top \mathbf{1}_n),$$

and, when Γ_n is invertible,

$$X_{n+h}^n = \mu + \gamma_{[h]}^\top \Gamma_n^{-1} (X - \mu \mathbf{1}_n), \quad P_{n+h}^n = \gamma_0 - \gamma_{[h]}^\top \Gamma_n^{-1} \gamma_{[h]}.$$

Proof idea: the predictor has the form $X_{n+h}^n = \mu + \beta^\top (X - \mu \mathbf{1}_n)$, so centring removes the mean and reduces the problem to the zero-mean case — there is no loss of generality in assuming $\mu = 0$, where $\beta_0 = 0$. The MSE is *identical* to the zero-mean case because the constant shift does not affect the error variance.

Why: real data are rarely centred; this is the version actually used in practice.

Theorem 10: Best linear predictor of a causal ARMA via ψ -weights

For a causal ARMA process with $MA(\infty)$ representation $X_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$, the best linear predictor of X_{n+h} based on $X_1, \dots, X_n$ is

$$X_{n+h}^n = \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j} = \psi_h \varepsilon_n + \psi_{h+1} \varepsilon_{n-1} + \dots,$$

with prediction mean square error

$$P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2.$$

Proof idea: write the predictor as a linear filter of the past innovations, $X_{n+h}^n = \sum_{j=0}^{\infty} c_j \varepsilon_{n-j}$. Then

$$\mathbb{E}[(X_{n+h} - X_{n+h}^n)^2] = \sigma^2 \left(\sum_{j=0}^{h-1} \psi_j^2 + \sum_{j=h}^{\infty} (\psi_j - c_{j-h})^2 \right),$$

which is minimised by matching $c_j = \psi_{j+h}$, killing the second sum and leaving $\sigma^2 \sum_{j=0}^{h-1} \psi_j^2$. Practically: **set the unrealised future innovations $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$ to zero** in $X_{n+h} = \sum_{j=0}^{\infty} \psi_j \varepsilon_{n+h-j}$.

Why: this is the engine of all ARMA forecasting. It also shows that for a causal ARMA, $\psi_j \rightarrow 0$ exponentially so $X_{n+h}^n \rightarrow \mu$, and $P_{n+h}^n = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \uparrow \sigma^2 \sum_{j=0}^{\infty} \psi_j^2 = \gamma_0$ as $h \rightarrow \infty$.

Proposition 11: Limiting behaviour of ARMA forecasts

For a causal ARMA model with mean μ , $X_{n+h}^n = \mu + \sum_{j=h}^{\infty} \psi_j \varepsilon_{n+h-j}$ with $\psi_j \rightarrow 0$ exponentially fast, hence

$$X_{n+h}^n \rightarrow \mu, \quad \text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 \rightarrow \gamma_0, \quad h \rightarrow \infty.$$

Proof idea: immediate from the ψ -weight theorem and exponential decay of ψ_j (a consequence of causality, roots of Φ outside the unit circle).

Why: quantifies “the distant future is the unconditional distribution” — the point forecast converges to the mean and the error variance converges to the marginal variance γ_0 .

Proposition 12: Forecasts for integrated (ARIMA) processes

Let $\{X_t\}$ be $ARIMA(p, d, q)$, $\Phi(B)(1-B)^d X_t = \Theta(B)\varepsilon_t$. Then $Z_t = \nabla^d X_t$ is the $ARMA(p, q)$ process $Z_t = \sum_{j=1}^p \phi_j Z_{t-j} + \varepsilon_t - \sum_{j=1}^q \theta_j \varepsilon_{t-j}$, and one forecasts X by undoing the differencing,

$$\nabla^d X_{n+h}^n = Z_{n+h}^n.$$

For $d = 1$ (so $X_{n+h} - X_{n+h-1} = Z_{n+h}$),

$$X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n, \quad h \geq 1.$$

If $\{Z_t\}$ has mean $\alpha \neq 0$ then $Z_{n+h}^n \rightarrow \alpha$ and $X_{n+h}^n \approx X_n + \alpha h$ for large h (a deterministic drift line).

Proof idea: apply the linear projection operator $P_{X_1, \dots, X_n}$, which is linear, to $\nabla^d X_{n+h} = Z_{n+h}$; since differencing is a fixed linear filter it commutes with the projection, giving $\nabla^d X_{n+h}^n = Z_{n+h}^n$. Telescoping the $d = 1$ identity $X_{n+h} = X_{n+h-1} + Z_{n+h}$ down to the last observation X_n yields

$$X_{n+h}^n = X_n + \sum_{j=0}^{h-1} Z_{n+h-j}^n.$$

Why: reduces ARIMA forecasting to ARMA forecasting on the differenced series, then “re-integrates”.

- For ARIMA models the prediction MSE **diverges**: $P_{n+h}^n \rightarrow \infty$ as $h \rightarrow \infty$ (no mean reversion — the process is non-stationary).
- For SARIMA models with $d = D = 1$ and $\alpha = 0$ the forecast is linear and seasonal in h .

9 Diagnostics, Model Selection and Long Memory

Once a candidate ARIMA model has been identified and fitted, two questions remain: *does it fit?* (diagnostics, based on the residuals) and *is it the right model among several?* (model selection, based on information criteria). This part collects the standard residual checks (mean, variance, Gaussianity, autocorrelation), the Box–Pierce and Ljung–Box portmanteau tests, the AIC/AICc/BIC criteria, and the Box–Jenkins methodology that ties them together. It then steps outside the ARMA world entirely: ARMA covariances decay exponentially fast, but many real series have covariances that decay only *polynomially*. This is **long memory**, modelled by fractional differencing $(1 - B)^d$ and the FARIMA class, and intimately related to self-similarity and fractional Brownian motion.

9.1 Definitions

Definition 1: Residuals and standardized residuals

For a fitted model with fitted value $\hat{X}_t$ at time t , the **residuals** are

$$e_t = X_t - \hat{X}_t,$$

and the **standardized residuals** are

$$\tilde{e}_t = \frac{e_t}{\sqrt{\text{Var}(e_t)}}.$$

Model checking is built almost entirely on the $\{e_t\}$ (or $\{\tilde{e}_t\}$): if the model is correct, the residuals should behave like the driving white noise. The four questions to ask are:

1. Do the residuals have constant zero mean?
2. Is their variance constant in t (*homoscedasticity*)?
3. Are they uncorrelated in t ?
4. Are they Gaussian?

Definition 2: Checking mean, variance and Gaussianity

Three of the four checks are visual:

- **Mean:** plot $\{(t, \tilde{e}_t)\}$. There should be no trend and the points should hover around zero.
- **Variance (homoscedasticity):** plot $\{(t, \tilde{e}_t)\}$ and look for constant spread; a fanning-out or fanning-in band signals non-constant variance.
- **Gaussianity:** a **Q–Q plot** of the residual quantiles against normal quantiles; Gaussian residuals lie on a straight line, whereas heavy tails (e.g. Student) curve away at the ends. Note this checks only *marginal* Gaussianity.

Definition 3: Sample residual autocorrelation & the correlogram band

The residual sample autocorrelation at lag τ is

$$\hat{r}_\tau = \hat{\rho}_\tau^{(e)} = \frac{\sum_{t=1}^{n-\tau} (e_{t+\tau} - \bar{e})(e_t - \bar{e})}{\sum_{t=1}^n (e_t - \bar{e})^2}.$$

Plotting $\hat{r}_\tau$ against τ gives the residual **correlogram**. Under the null hypothesis that the residuals are white noise, each $\hat{r}_\tau$ is approximately $\mathcal{N}(0, 1/n)$, so individual values are compared against the band

$$\left(-\frac{1.96}{\sqrt{n}}, \frac{1.96}{\sqrt{n}} \right),$$

the (pointwise) 95% acceptance band for “ $\hat{r}_\tau$ is zero”.

Definition 4: R^2 statistic

With s_e^2 the residual variance and s_y^2 the sample variance of the data,

$$R^2 = 1 - \frac{s_e^2}{s_y^2}.$$

It measures the fraction of variance “explained” by the fit. In time-series model selection it is a **dangerous** criterion (see Proposition 13): adding parameters can only increase R^2 , and even the *true* model has a bounded R^2 because a stochastic process is not perfectly predictable.

Definition 5: Information criteria: AIC, AICc, BIC

For a model with k estimated parameters fitted to n observations, with maximized likelihood $L(\theta | y)$:

$$\text{AIC}(\theta) = -2 \log L(\theta | y) + 2k,$$

$$\text{AICc}(\theta) = -2 \log L(\theta | y) + 2k \frac{n}{n - k - 1} = \text{AIC}(\theta) + \frac{2k(k + 1)}{n - k - 1},$$

$$\text{BIC}(\theta) = -2 \log L(\theta | y) + k \log(n).$$

Smaller is better for all three. For an ARMA(p, q) the parameter count is

$$k = p + q + 1$$

(p AR coefficients, q MA coefficients, plus the noise variance σ^2). AIC is known to *overestimate* p ; AICc (Hurvich & Tsai, 1989) corrects the small-sample bias and coincides with AIC as $n \rightarrow \infty$; BIC penalizes parameters more heavily (for $n > e^2 \approx 7.4$) and so favours more parsimonious models.

Definition 6: Long memory (covariance definition)

A second-order stationary process $\{X_t\}$ with ACVS $\{\gamma_\tau\}$ is said to exhibit **long memory** (long-range dependence) if its covariances decay only polynomially:

$$\gamma_\tau \sim L_\gamma(\tau) |\tau|^{-\alpha}, \quad |\tau| \rightarrow \infty,$$

where L_γ is a *slowly varying* function and the exponent satisfies $0 < \alpha < 1$. Contrast this with ARMA models, whose covariances are either finite-support ($\gamma_\tau = 0$ for $|\tau| > q$) or exponentially decaying ($\gamma_\tau \sim |\phi|^\tau$). The full admissible range $0 < \alpha < 2$ yields three regimes:

$$\underbrace{0 < \alpha < 1}_{\text{long memory}}, \quad \underbrace{\alpha = 1}_{\text{short-range dependence}}, \quad \underbrace{1 < \alpha < 2}_{\text{anti-persistence}}.$$

Definition 7: Long memory (spectral definition)

Equivalently, long memory is characterized by a spectral density that blows up (or vanishes) like a power of the frequency near the origin:

$$\mathcal{S}(f) \sim L_S(f) |f|^{\alpha-1}, \quad f \rightarrow 0,$$

with L_S slowly varying. For $0 < \alpha < 1$ the exponent $\alpha - 1 \in (-1, 0)$, so $\mathcal{S}(f) \rightarrow \infty$ as $f \rightarrow 0$ (a spectral pole at zero): the process concentrates power at the lowest frequencies, the spectral signature of persistence. The slowly varying parts of the two definitions are linked by

$$L_S(f) = 2 \gamma(0) \Gamma(1 - \alpha) \sin \frac{\pi \alpha}{2}, \quad L_\gamma(\tau) = 2 \Gamma(\alpha) \sin \frac{\pi(1 - \alpha)}{2}.$$

The case $\alpha = 1$ (short-range dependence) gives $\mathcal{S}(0)$ finite and nonzero, exactly the ARMA situation $\mathcal{S}(0) = \sum_\tau \gamma_\tau < \infty$; the anti-persistent case $1 < \alpha < 2$ forces $\sum_k \gamma_k = 0$ (and $\mathcal{S}(0) = 0$).

Definition 8: Fractional differencing operator & FARIMA

The **fractional differencing operator** is defined for real d via the binomial expansion of $(1 - B)^d$ in the backshift operator B :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-1)^k B^k = \sum_{k=0}^{\infty} \frac{\Gamma(d+1)}{\Gamma(k+1) \Gamma(d-k+1)} (-1)^k B^k = \sum_{j=0}^{\infty} \frac{\Gamma(j-d)}{\Gamma(-d) \Gamma(j+1)} B^j.$$

The simplest long-memory process is the **fractionally differenced** (pure) process $(1 - B)^d X_t = \varepsilon_t$. The general **FARIMA** (fractional ARIMA) class is the stationary solution of

$$(1 - B)^d \phi(B) X_t = \psi(B) \varepsilon_t, \quad d = \frac{1 - \alpha}{2} \in (-0.5, 0.5),$$

where ε_t are i.i.d. zero-mean with variance $\sigma_\varepsilon^2 > 0$, and $\phi(z)$, $\psi(z)$ are degree- p and degree- q polynomials with all roots outside the unit circle. Integer d recovers ordinary ARIMA; fractional $d \in (0, 0.5)$ gives long memory.

Definition 9: Self-similarity & fractional Brownian motion

A process $\{Y_t\}$ is **self-similar** with self-similarity parameter (Hurst exponent) H if, for every $c > 0$,

$$Y_{ct} \stackrel{d}{=} c^H Y_t$$

(equality in distribution, jointly over t). Self-similarity with stationary increments forces the covariance structure

$$\text{Cov}(Y_t, Y_s) = \frac{\sigma^2}{2} \left(|t|^{2H} + |s|^{2H} - |t - s|^{2H} \right). \quad (\star)$$

A *Gaussian* process with covariance $(\star)$ is **fractional Brownian motion** $B_H(t)$; it is the *only* self-similar Gaussian process. Its increment process $X_t = B_H(t) - B_H(t - 1)$ is **fractional Gaussian noise**, with

$$\gamma_X(\tau) = \frac{\sigma^2}{2} \left(|\tau + 1|^{2H} - 2|\tau|^{2H} + |\tau - 1|^{2H} \right) \sim H(2H - 1) |\tau|^{2H-2}, \quad |\tau| \rightarrow \infty.$$

The graph $(t, B_H(t))$ has Hausdorff (fractal) dimension $D = 2 - H$ in one dimension. Fractal dimension is a *local* property; long-range dependence is a *global* one — the two can be made completely independent.

9.2 Theorems & Propositions

Proposition 10: Residuals of a correctly specified AR(1) are the noise

For an AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$ with known true parameters, the fitted one-step value is $\hat{X}_t = \phi X_{t-1}$, so the residual is

$$e_t = X_t - \phi X_{t-1} = \varepsilon_t.$$

Proof idea: substitute the model equation; the AR structure cancels exactly, leaving the innovation.

Why: this is the template for *all* residual diagnostics. If the model and parameters are right, residuals reproduce the white noise: constant zero mean, constant variance, uncorrelated, and Gaussian whenever ε_t is. Any departure from these is evidence *against* the model.

Proposition 11: Box–Pierce portmanteau statistic

To test $H_0 : \rho_1 = \rho_2 = \dots = \rho_m = 0$ (residuals white up to lag m) against the alternative that some $\rho_\tau \neq 0$, the **Box–Pierce** statistic is

$$Q_m = n \sum_{\tau=1}^m \hat{r}_\tau^2.$$

Under H_0 , for an ARMA(p, q) fit, Q_m is approximately χ_{m-p-q}^2 .

Proof idea: under white noise each $\sqrt{n} \hat{r}_\tau \xrightarrow{d} \mathcal{N}(0, 1)$ and the $\hat{r}_\tau$ are asymptotically independent, so $\sum_{\tau} (\sqrt{n} \hat{r}_\tau)^2$ is a sum of squared standard normals; fitting $p + q$ parameters removes $p + q$ degrees of freedom.

Why: it tests *many* lags at once (a “portmanteau”), avoiding the multiple-testing problem of eyeballing the correlogram lag by lag.

Proposition 12: Ljung–Box statistic (modified Box–Pierce)

The small-sample-corrected (**Ljung–Box**) statistic is

$$Q_m = n(n+2) \sum_{\tau=1}^m \frac{\hat{r}_\tau^2}{n-\tau},$$

again approximately χ_{m-p-q}^2 under H_0 for an ARMA(p, q).

Proof idea: the factor $\frac{n+2}{n-\tau}$ rescales each term so its finite- n variance matches the χ^2 target more closely than the plain $n\hat{r}_\tau^2$; as $n \rightarrow \infty$, $\frac{n(n+2)}{n-\tau} \rightarrow n$ and the two statistics agree.

Why: the Box–Pierce χ^2 approximation is poor at realistic sample sizes, so Ljung–Box is the version used in practice (e.g. R’s `Box.test(type="Ljung-Box")`). One usually computes Q_m over a range of m and plots the resulting p -values.

Proposition 13: The danger of R^2 : an AR(1) caps at ϕ^2

For an AR(1) with innovation variance σ^2 , the process variance is $s_y^2 = \sigma^2/(1 - \phi^2)$ and the one-step residual variance is $s_e^2 = \sigma^2$, so

$$R^2 = 1 - \frac{s_e^2}{s_y^2} = 1 - \frac{\sigma^2(1 - \phi^2)}{\sigma^2} = \phi^2.$$

Proof idea: substitute the AR(1) variance formula (geometric sum of the MA(∞) coefficients ϕ^{2j}) and the fact that the best one-step prediction error is ε_t .

Why: even the *true* model has bounded R^2 ($\phi = 0.4 \Rightarrow R^2 = 0.16$, which “looks bad” but is

optimal). A stochastic process is not perfectly predictable, so a low R^2 is no indictment, and a high one (from piling on parameters) is no endorsement. **Use information criteria, not R^2 .**

Proposition 14: AIC vs. AICc gap

The corrected and uncorrected AIC differ by exactly the parameter-count penalty

$$\text{AICc}(\theta) - \text{AIC}(\theta) = 2k \left(\frac{n}{n-k-1} - 1 \right) = \frac{2k(k+1)}{n-k-1} (> 0).$$

Proof idea: algebra — put $\frac{n}{n-k-1} - 1 = \frac{k+1}{n-k-1}$ over a common denominator.

Why: the correction is always positive (it penalizes complexity more), vanishes as $n \rightarrow \infty$ with k fixed, and stays negligible whenever $k^2/n \rightarrow 0$. AICc is therefore the safe default in small samples or when k is comparable to n .

Proposition 15: BIC penalty exceeds AIC penalty for $n > e^2$

The per-parameter penalties are 2 (AIC) and $\log n$ (BIC). Hence the BIC penalty is the larger of the two iff

$$\log n > 2 \iff n > e^2 \approx 7.39.$$

Proof idea: compare the per-parameter coefficients 2 and $\log n$ directly.

Why: for any realistic n (≥ 8) BIC charges more per parameter, hence is more conservative and consistent (selects the true order with probability $\rightarrow 1$), whereas AIC is more permissive and tends to over-fit p . Choose BIC when you want parsimony / consistency, AIC(c) for predictive performance.

Proposition 16: Long-memory inflation of $\text{Var}(\bar{X})$

For a stationary process with absolutely continuous spectrum, $\text{Var}(\bar{X}) \sim \mathcal{S}(0)/n$ (the classical $1/n$ rate, since $\mathcal{S}(0) = \sum_{\tau} \gamma_{\tau} < \infty$). Under long memory with $\gamma_{\tau} \sim |\tau|^{-\alpha}$ ($0 < \alpha < 1$), however,

$$\text{Var}(\bar{X}) \sim \frac{c}{n^{\alpha}}, \quad 0 < \alpha < 1,$$

i.e. the variance decays *slower* than $1/n$.

Proof idea: $\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i,j} \gamma_{j-i}$; aggregating $\sum_{j=-n}^n \gamma_j$ with $\gamma_j \sim |j|^{-\alpha}$ recovers a power $n^{1-\alpha}$, so dividing by n gives $n^{-\alpha}$.

Why: this is the empirical fingerprint of long memory (Smith, 1938): sample means converge much more slowly than i.i.d. or short-range theory predicts, so naive standard errors $\sqrt{\mathcal{S}(0)/n}$ are badly too small.

Theorem 17: Spectral density of a FARIMA / fractionally differenced process

The spectral density of the FARIMA process $(1-B)^d \phi(B)X_t = \psi(B)\varepsilon_t$ is

$$\mathcal{S}(f) = \sigma_{\varepsilon}^2 \frac{|\psi(e^{-2\pi i f})|^2}{|\phi(e^{-2\pi i f})|^2} \left| 1 - e^{-2\pi i f} \right|^{-2d} \underset{f \rightarrow 0}{\sim} c_f |f|^{-2d}, \quad c_f = \sigma_{\varepsilon}^2 \frac{|\psi(1)|^2}{|\phi(1)|^2}.$$

Proof idea: the spectral density of a filtered process multiplies the input spectrum by the squared transfer function; the fractional difference contributes $|1 - e^{-2\pi i f}|^{-2d}$, and near $f = 0$, $|1 - e^{-2\pi i f}| \sim 2\pi |f|$, leaving the pole $|f|^{-2d}$.

Why: it shows the long-memory parameter α ($d = (1 - \alpha)/2$, so $-2d = \alpha - 1$) is read off the slope of $\log \mathcal{S}$ vs. $\log |f|$ near the origin — the basis of log-periodogram (GPH) estimation of d .

Proposition 18: Estimating d from the periodogram (log-periodogram idea)

Since $\mathbb{E}I(f) \approx \mathcal{S}(f) \sim L_S(f)|f|^{\alpha-1}$ near $f = 0$, taking logarithms gives, for low Fourier frequencies f_j ,

$$\log I(f_j) \approx \text{const} + (\alpha - 1) \log |f_j| + \text{error} = \text{const} - 2d \log |f_j| + \text{error}.$$

Proof idea: the periodogram is an (asymptotically unbiased but inconsistent) estimate of the spectral density; regressing $\log I(f_j)$ on $\log |f_j|$ over the lowest frequencies estimates the slope $\alpha - 1 = -2d$.

Why: it turns long-memory estimation into a simple linear regression on the log-log periodogram; one must take log *before* taking expectations, because $\mathbb{E} \log I \neq \log \mathbb{E} I$.

10 Non-Linear Time Series: ARCH and GARCH

Linear models (AR, MA, ARMA) describe the conditional *mean* of a series and assume a constant innovation variance. Financial log-returns violate this: they are nearly uncorrelated (so a linear conditional-mean model is almost useless), yet their *squares* are strongly dependent — large moves cluster with large moves and quiet periods with quiet periods. This **volatility clustering**, together with heavy (**fat**) tails, motivates models for the *conditional variance*. **ARCH** and **GARCH** keep the series mean-zero and uncorrelated (white noise) while letting the conditional variance σ_t^2 evolve in time as a function of past squared returns (and, for GARCH, past variances). Throughout, $\mathcal{F}_t$ denotes the information available up to and including time t , and $\{\varepsilon_t\}$ is white noise with $\mathbb{E}[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = 1$, independent of $\mathcal{F}_{t-1}$.

10.1 Definitions

Definition 1: Log-returns & volatility clustering

For a price series $\{X_t\}$ (e.g. a stock index), the **log-returns** are

$$r_t = \log\left(\frac{X_t}{X_{t-1}}\right) = \log(X_t) - \log(X_{t-1}),$$

measuring relative changes in price. Empirically log-returns show:

- **little linear dependence:** the sample ACF/PACF of r_t is essentially that of white noise;
- **dependent squares:** the sample ACF/PACF of r_t^2 decays slowly $\Rightarrow$ the conditional variance changes over time;
- **volatility clustering:** large changes tend to be followed by large changes, small by small;
- **fat tails:** kurtosis > 3 (heavier tails than the normal), visible as deviations in a QQ-plot against $\mathcal{N}(0, 1)$.

Definition 2: ARCH(1) model

$\{r_t\}$ is an **AutoRegressive Conditionally Heteroscedastic** model of order 1, ARCH(1), if for $t \in \mathbb{Z}$

$$r_t = \sigma_t \varepsilon_t, \quad \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2,$$

with $\{\varepsilon_t\}$ white noise of mean 0 and variance 1, independent of $\mathcal{F}_{t-1}$, and parameters $\alpha_0 > 0$, $\alpha_1 \geq 0$. The volatility σ_t is $\mathcal{F}_{t-1}$ -measurable (known one step ahead). Immediate properties (Proposition 7):

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0, \quad \text{Var}(r_t \mid \mathcal{F}_{t-1}) = \sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2.$$

Definition 3: ARCH(p) model

$\{r_t\}$ is an ARCH(p) model if $r_t = \sigma_t \varepsilon_t$ with

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2 + \cdots + \alpha_p r_{t-p}^2 = \alpha_0 + \sum_{j=1}^p \alpha_j r_{t-j}^2,$$

$\{\varepsilon_t\}$ white noise $(0, 1)$ independent of $\mathcal{F}_{t-1}$, and $\alpha_0 > 0$, $\alpha_p > 0$, $\alpha_j \geq 0$ for $1 \leq j < p$. Volatility may now depend on several past squared shocks, so as before $\text{Var}(r_t | \mathcal{F}_{t-1}) = \sigma_t^2$.

Definition 4: GARCH(p,q) model

$\{r_t\}$ is a **Generalised ARCH** model GARCH(p, q) if $r_t = \sigma_t \varepsilon_t$ with

$$\sigma_t^2 = \alpha_0 + \sum_{j=1}^p \alpha_j r_{t-j}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

$\{\varepsilon_t\}$ white noise $(0, 1)$ independent of $\mathcal{F}_{t-1}$, and parameters $\alpha_0 > 0$, $\alpha_p > 0$, $\beta_q > 0$, with $\alpha_j, \beta_j \geq 0$ otherwise. Feeding back past *variances* σ_{t-j}^2 lets volatility persist over time far more parsimoniously than an ARCH(p) with large p . A GARCH($p, 0$) is exactly an ARCH(p). As before $\text{Var}(r_t | \mathcal{F}_{t-1}) = \sigma_t^2$.

Definition 5: Standardised residuals (diagnostics)

After fitting, the **standardised residuals** are $\tilde{e}_t = r_t / \hat{\sigma}_t$. If the model is correct these should look like i.i.d. $(0, 1)$ noise: their ACF/PACF *and* the ACF/PACF of their *squares* $\tilde{e}_t^2$ should resemble white noise, and a QQ-plot against the assumed innovation law should be roughly straight. Under Gaussian innovations one also overlays in-sample bands $\pm 1.96 \hat{\sigma}_t$ on r_t .

Definition 6: ARMA–GARCH model

Because a GARCH process is itself white noise, it can serve as the innovation of an ARMA model: with r_t a GARCH($\tilde{p}, \tilde{q}$) process,

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} - \sum_{k=0}^q \theta_k r_{t-k}, \quad \theta_0 = -1,$$

where the ϕ_j, θ_k satisfy the usual ARMA(p, q) stationarity and invertibility conditions. This captures a conditional mean (ARMA) and a time-varying conditional variance (GARCH) at once; one must only be careful with the likelihood, as the errors are no longer i.i.d. Gaussian.

10.2 Theorems & Propositions

Proposition 7: Conditional and unconditional moments of ARCH(1)

For the ARCH(1) model with $\alpha_1 < 1$:

$$\mathbb{E}[r_t | \mathcal{F}_{t-1}] = 0, \quad \text{Var}(r_t | \mathcal{F}_{t-1}) = \sigma_t^2, \quad \mathbb{E}[r_t] = 0, \quad \text{Var}(r_t) = \frac{\alpha_0}{1 - \alpha_1}.$$

Proof idea: σ_t is $\mathcal{F}_{t-1}$ -measurable and ε_t is independent of $\mathcal{F}_{t-1}$, so $\mathbb{E}[r_t | \mathcal{F}_{t-1}] = \sigma_t \mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}[r_t^2 | \mathcal{F}_{t-1}] = \sigma_t^2 \mathbb{E}[\varepsilon_t^2] = \sigma_t^2$. Iterated expectation gives $\mathbb{E}[r_t] = \mathbb{E}[\mathbb{E}[r_t | \mathcal{F}_{t-1}]] = 0$ and $\text{Var}(r_t) = \mathbb{E}[\sigma_t^2] = \alpha_0 + \alpha_1 \text{Var}(r_{t-1})$; under stationarity $\text{Var}(r_t) = \text{Var}(r_{t-1})$, solve to get $\alpha_0/(1 - \alpha_1)$.

Why: establishes that ARCH is mean-zero with a positive finite unconditional variance exactly when $\alpha_1 < 1$ (necessary *and*, for ARCH(1), sufficient for weak stationarity).

Proposition 8: ARCH/GARCH is white noise (uncorrelated but dependent)

Any ARCH(p) or GARCH(p, q) process is mean-zero and serially uncorrelated: for $\tau \neq 0$,

$$\mathbb{E}[r_t] = 0, \quad \text{Cov}(r_{t+\tau}, r_t) = 0, \quad \text{hence} \quad \text{Corr}(r_{t+\tau}, r_t) = 0.$$

Proof idea: take $\tau > 0$. Since r_t is $\mathcal{F}_{t+\tau-1}$ -measurable and $\sigma_{t+\tau}$ is too, condition on $\mathcal{F}_{t+\tau-1}$:

$$\text{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t] = \mathbb{E}[r_t \sigma_{t+\tau} \mathbb{E}[\varepsilon_{t+\tau} | \mathcal{F}_{t+\tau-1}]] = \mathbb{E}[r_t \sigma_{t+\tau} \cdot 0] = 0.$$

Why: the central qualitative fact — ARCH/GARCH leaves the *linear* correlation structure flat (white noise) while creating strong *nonlinear* dependence in the squares. The conditional-mean and conditional-variance modelling problems decouple.

Theorem 9: Isserlis' theorem (Wick formula)

If $(X_1, \dots, X_N)$ is a zero-mean multivariate normal vector and N is even,

$$\mathbb{E}[X_1 \cdots X_N] = \sum_{p \in P_N} \prod_{\{i, j\} \in p} \text{Cov}(X_i, X_j),$$

where P_N is the set of all pairings (perfect matchings) of $\{1, \dots, N\}$. For $N = 4$,

$$\mathbb{E}[X_1 X_2 X_3 X_4] = \mathbb{E}[X_1 X_2] \mathbb{E}[X_3 X_4] + \mathbb{E}[X_1 X_3] \mathbb{E}[X_2 X_4] + \mathbb{E}[X_1 X_4] \mathbb{E}[X_2 X_3].$$

Proof idea: expand the Gaussian moment generating function / cumulant expansion; only pair-cumulants (covariances) survive for a Gaussian.

Why: gives $\mathbb{E}[\varepsilon_t^4] = 3(\mathbb{E}[\varepsilon_t^2])^2 = 3$ for standard Gaussian innovations, the key input to the ARCH fourth-moment and kurtosis computations below.

Proposition 10: Fourth moment and kurtosis of ARCH(1)

For ARCH(1) with Gaussian innovations, the unconditional fourth moment is finite iff $0 \leq \alpha_1^2 < \frac{1}{3}$, in which case

$$\mathbb{E}[r_t^4] = \frac{3\alpha_0^2(1 + \alpha_1)}{(1 - \alpha_1)(1 - 3\alpha_1^2)}, \quad \kappa = \frac{\mathbb{E}[r_t^4]}{\text{Var}(r_t)^2} = \frac{3(1 - \alpha_1^2)}{1 - 3\alpha_1^2} > 3 \quad (\alpha_1 > 0).$$

Proof idea: $\mathbb{E}[r_t^4 | \mathcal{F}_{t-1}] = \sigma_t^4 \mathbb{E}[\varepsilon_t^4 | \mathcal{F}_{t-1}] = 3\sigma_t^4 = 3(\alpha_0 + \alpha_1 r_{t-1}^2)^2$ (Isserlis). Take expectations, substitute $\mathbb{E}[r_{t-1}^2] = \alpha_0/(1 - \alpha_1)$ and $\mathbb{E}[r_{t-1}^4] = \mathbb{E}[r_t^4]$ by stationarity, and solve the linear equation; divide by $\text{Var}(r_t)^2 = \alpha_0^2/(1 - \alpha_1)^2$ for κ .

Why: shows ARCH *generates* excess kurtosis (fat tails) endogenously, even with Gaussian ε_t — matching the heavy tails seen in returns.

Proposition 11: ACF of the squares of an ARCH(1)

For ARCH(1) with $0 < \alpha_0$ and $0 < \alpha_1 < 1/\sqrt{3}$,

$$\text{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}, \quad \tau \in \mathbb{Z},$$

i.e. the autocorrelation of the squared series decays *geometrically*.

Proof idea: condition on $\mathcal{F}_{t+\tau-1}$ to peel one step: $\mathbb{E}[r_{t+\tau}^2 r_t^2 \mid \mathcal{F}_{t+\tau-1}] = r_t^2 \sigma_{t+\tau}^2 = r_t^2(\alpha_0 + \alpha_1 r_{t+\tau-1}^2)$. Taking expectations and subtracting $(\mathbb{E}[r_t^2])^2$ yields the recursion $\text{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1 \text{Cov}(r_{t+\tau-1}^2, r_t^2)$ (the constant terms cancel using $\mathbb{E}[r_t^2] = \alpha_0/(1 - \alpha_1)$). Iterating gives $\text{Cov}(r_{t+\tau}^2, r_t^2) = \alpha_1^\tau \text{Var}(r_t^2)$, hence the stated correlation.

Why: explains the slowly decaying ACF of squared returns and warns that ARCH(1) imposes *purely geometric* memory in volatility — often too fast for real data, motivating GARCH.

Proposition 12: Stationarity & variance of GARCH(p,q)

A GARCH(p, q) process has a weakly stationary solution **iff**

$$\sum_{j=1}^p \alpha_j + \sum_{j=1}^q \beta_j < 1,$$

and then its unconditional variance is

$$\text{Var}(r_t) = \frac{\alpha_0}{1 - \sum_{j=1}^p \alpha_j - \sum_{j=1}^q \beta_j}.$$

Proof idea: take unconditional expectations of $\sigma_t^2 = \alpha_0 + \sum_j \alpha_j r_{t-j}^2 + \sum_j \beta_j \sigma_{t-j}^2$ using $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2] = \text{Var}(r_t)$ at stationarity, then solve; positivity of the denominator forces the stated bound.

Why: the master condition for fitting GARCH — it must hold for the fitted parameters, and the closed-form variance gives the long-run volatility level. (More: $\sum \alpha + \sum \beta < 1$ is sufficient for strict stationarity; for some noise laws including Gaussian, the boundary $\sum \alpha + \sum \beta = 1$ still admits a strictly — but not weakly — stationary solution: the IGARCH case.)

Proposition 13: Fourth-moment condition for GARCH(1,1)

For a GARCH(1, 1) with Gaussian noise, the unconditional fourth moment $\mathbb{E}[r_t^4]$ exists **iff**

$$3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2 < 1.$$

Proof idea: square the recursion for σ_t^2 , take expectations using $\mathbb{E}[\varepsilon_t^4] = 3$ (Isserlis) and stationarity of $\mathbb{E}[\sigma_t^4]$; the coefficient of $\mathbb{E}[\sigma_t^4]$ on the right is $3\alpha_1^2 + 2\alpha_1\beta_1 + \beta_1^2$, which must be < 1 for a finite solution.

Why: the analogue of $\alpha_1^2 < 1/3$; needed before quoting any formula for the kurtosis or the ACF of squared GARCH returns.

Proposition 14: Maximum likelihood estimation (conditional Gaussian)

Conditioning on the first observation r_1 to avoid specifying a marginal law for the initial value, the conditional likelihood factorises as

$$L_c(\theta \mid r_1, \dots, r_n) = \prod_{t=2}^n f(r_t \mid r_{t-1}, \dots, r_1; \theta).$$

Under Gaussian innovations the ARCH(1) model gives $r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, \alpha_0 + \alpha_1 r_{t-1}^2)$, so

$$L_c(\alpha_0, \alpha_1) = \prod_{t=2}^n \frac{1}{\sqrt{2\pi(\alpha_0 + \alpha_1 r_{t-1}^2)}} \exp\left(-\frac{r_t^2}{2(\alpha_0 + \alpha_1 r_{t-1}^2)}\right),$$

maximised over $\alpha_0 > 0, \alpha_1 \geq 0$ (numerically; GARCH is analogous with σ_t^2 computed by its recursion).

Proof idea: the chain rule for densities plus conditional normality of each r_t given the past.

Why: the standard estimation route for ARCH/GARCH; the per-term variance σ_t^2 is what is being fitted, not a constant σ^2 .

11 Linear Time-Invariant (LTI) Filters

A (*digital*) *filter* is a map that turns one sequence into another, $\{Y_t\} = \mathcal{L}[\{X_t\}]$, with index set $\mathbb{Z}$. The filters that matter for time series are the **linear time-invariant** (LTI) ones: they treat all time points alike and act linearly. Two “test inputs” completely characterise such a filter — a single impulse (giving the *impulse response* h) and a pure complex wave (giving the *transfer function* H). Their power is twofold: every LTI filter is a *convolution* with h , and in the frequency domain it acts *diagonally*, multiplying the spectral density by $|H(f)|^2$. This single fact, $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$, is the engine behind the closed-form spectral density of any ARMA process. Throughout we allow time series to be *complex-valued* and assume the (possibly infinite) sums involved converge, the standard sufficient condition being $h \in \ell^1$.

11.1 Definitions

Definition 1: Complex-valued time series & its autocovariance

A **complex-valued** time series can be written $X_t = U_t + iV_t$ with $\{U_t\}, \{V_t\}$ real-valued, and has mean $\mathbb{E}[X_t] = \mathbb{E}[U_t] + i\mathbb{E}[V_t]$. Its **autocovariance** uses a *conjugate* on the second factor:

$$\gamma_\tau^{(X)} = \text{Cov}(X_{t+\tau}, X_t) = \mathbb{E}[(X_{t+\tau} - \mu)(\overline{X_t - \mu})].$$

This reduces to the usual real ACVS when $X_t \in \mathbb{R}$. (Stationarity is required jointly of the real and imaginary parts.) Throughout we drop the index set: $\{X_t\} = \{X_t\}_{t \in \mathbb{Z}}$.

Definition 2: Digital filter & output notation

A **digital filter** $\mathcal{L}$ maps an input sequence to an output sequence, $\{Y_t\} = \mathcal{L}[\{X_t\}]$. To name a single output entry we put the time index *after* the bracket,

$$Y_u = \mathcal{L}[\{X_t\}]_u, \quad u \in \mathbb{Z}.$$

The t in $\{X_t\}$ is a *dummy* variable; the $u \in \mathbb{Z}$ after the bracket has a definite meaning (the output time).

Definition 3: Linear time-invariant (LTI) filter

A digital filter $\mathcal{L}$ is **linear time-invariant** if for all sequences $\{X_t\}, \{Y_t\}$ and all $\alpha \in \mathbb{C}$:

1. **Linearity:** $\mathcal{L}[\{\alpha X_t + Y_t\}] = \alpha \mathcal{L}[\{X_t\}] + \mathcal{L}[\{Y_t\}]$;
2. **Time invariance:** $\mathcal{L}[B[\{X_t\}]] = B[\mathcal{L}[\{X_t\}]]$, i.e. $\mathcal{L}$ commutes with the backshift, $\mathcal{L}B = B\mathcal{L}$.

Time invariance for the single shift B already implies it for every shift: $\mathcal{L}[B^u\{X_t\}] = B^u\mathcal{L}[\{X_t\}]$ for all $u \in \mathbb{Z}$. (The continuous-time analogue is usually stated with B^u , but the weaker one-step condition suffices.)

Definition 4: Impulse sequence & impulse response

For $m \in \mathbb{Z}$ the **impulse sequence** $\{\delta_{t,m}\}$ is $\delta_{t,m} = 1$ if $t = m$ and 0 otherwise. The **impulse response** of an LTI filter $\mathcal{L}$ is the sequence $\{h_m\}$ defined by

$$h_m = \mathcal{L}[\{\delta_{t,-m}\}]_0, \quad m \in \mathbb{Z},$$

where the subscript 0 is the *output time index*. Informally h_m is the response, read off at time 0, to a unit shock placed at time $-m$. A common sufficient regularity condition is $h \in \ell^1$ (then $\|h\|_\infty \leq \|h\|_1 < \infty$).

Definition 5: Complex wave & transfer function

For $f \in \mathbb{R}$ the **complex wave** is the sequence $\{\xi_{t,f}\}$ with $\xi_{t,f} = e^{2\pi i f t}$. The **transfer function** of an LTI filter $\mathcal{L}$ that can be applied to these waves is

$$H(f) = \mathcal{L}[\{\xi_{t,f}\}]_0, \quad f \in \mathbb{R}.$$

It encodes the frequency-domain behaviour of $\mathcal{L}$: how each pure frequency is re-weighted in amplitude and shifted in phase.

11.2 Theorems & Propositions

Proposition 6: Linear combination of LTI filters is LTI

If $\mathcal{L}_1, \mathcal{L}_2$ are LTI and $\alpha \in \mathbb{C}$, then $\mathcal{L} = \alpha\mathcal{L}_1 + \mathcal{L}_2$ is LTI.

Proof idea: apply linearity of $\mathcal{L}_1, \mathcal{L}_2$ to $\mathcal{L}[\beta\{X_t\} + \{Y_t\}]$ and regroup; for time invariance use that each $\mathcal{L}_j$ commutes with B and that B is linear, so $\mathcal{L}[B\{X_t\}] = B[\mathcal{L}\{X_t\}]$.

Why: it makes the LTI filters a *vector space* (over $\mathbb{C}$); in particular any polynomial in B , e.g. $\Phi(B) = \sum_j \phi_j B^j$, is LTI.

Proposition 7: Cascade of LTI filters is LTI

If $\mathcal{L}_1, \mathcal{L}_2$ are LTI, then the **cascade** $\mathcal{L} = \mathcal{L}_1\mathcal{L}_2$, defined by $\mathcal{L}[\{X_t\}] = \mathcal{L}_1[\mathcal{L}_2[\{X_t\}]]$, is LTI.

Proof idea: linearity is the composition of two linear maps; time invariance follows by pushing B through both filters, $\mathcal{L}_1[\mathcal{L}_2[B\{X_t\}]] = \mathcal{L}_1[B[\mathcal{L}_2\{X_t\}]] = B[\mathcal{L}_1[\mathcal{L}_2\{X_t\}]]$.

Why: filters can be *chained* and the family is closed under composition; combined with the previous proposition, all of $\{\text{polynomials and power series in } B\}$ are LTI.

Theorem 8: LTI $\Leftrightarrow$ convolution with the impulse response

Under suitable convergence assumptions, an LTI filter with impulse response h acts as a **convolution**:

$$\mathcal{L}[\{X_t\}]_u \stackrel{\text{ms}}{=} \sum_{m \in \mathbb{Z}} h_{u-m} X_m, \quad u \in \mathbb{Z}.$$

Conversely, any filter defined by such a convolution (with, say, $h \in \ell^1$) is LTI. The impulse response gives the convolution weights.

Proof idea (deterministic ℓ^1 version): write $X^{(N)} = \sum_{|m| \leq N} X_m \delta_{.,m}$; by continuity in ℓ^1 , $\mathcal{L}[\{X_t\}]_u = \lim_N \mathcal{L}[X^{(N)}]_u$. Linearity turns each finite truncation into a finite sum, and time

invariance gives $\mathcal{L}[\{\delta_{t,m}\}]_u = \mathcal{L}[\{\delta_{t,m-u}\}]_0 = h_{u-m}$; absolute convergence ($h \in \ell^1$, $X \in \ell^1$) lets the partial sums pass to the full series. The converse checks linearity and time invariance directly on the convolution (re-indexing $m' = m - 1$ for the shift).

Why: this is the master representation — it reduces *any* LTI filter to a single object, the weights h , and underlies every spectral computation below.

Theorem 9: Complex waves are eigensequences; $H(f)$ is the eigenvalue

Let $\mathcal{L}$ be an LTI filter applicable to the complex waves $\{\xi_{t,f}\}$. Then each wave is an **eigensequence**: for every $f \in \mathbb{R}$,

$$\mathcal{L}[\{\xi_{t,f}\}] = H(f) \{\xi_{t,f}\},$$

i.e. $\mathcal{L}$ only rescales amplitude and shifts phase while preserving the frequency.

Proof idea: since $B^{-u}[\{\xi_{t,f}\}]_\tau = \xi_{\tau+u,f} = e^{2\pi i u f} \xi_{\tau,f}$, we have $B^{-u}[\{\xi_{t,f}\}] = e^{2\pi i u f} \{\xi_{t,f}\}$. Then for any u ,

$$\mathcal{L}[\{\xi_{t,f}\}]_u = B^u[\mathcal{L}[B^{-u}\{\xi_{t,f}\}]]_u = \mathcal{L}[e^{2\pi i f u} \{\xi_{t,f}\}]_0 = e^{2\pi i f u} \mathcal{L}[\{\xi_{t,f}\}]_0 = \xi_{u,f} H(f),$$

using time invariance, the shift identity, linearity and the definition of H .

Why: it diagonalises every LTI filter in the frequency basis — filtering becomes *pointwise multiplication* by $H(f)$, the source of all clean frequency-domain formulas.

Theorem 10: Transfer function = Fourier transform of h

If the impulse response satisfies $h \in \ell^1$, the transfer function is the (discrete) Fourier transform of h :

$$H(f) = \sum_{m \in \mathbb{Z}} h_m e^{-2\pi i f m}, \quad f \in \mathbb{R}.$$

Proof idea: feed the wave $\xi_{t,f}$ into the convolution representation: $H(f) = \mathcal{L}[\{\xi_{t,f}\}]_0 = \sum_m h_{-m} e^{2\pi i f m} = \sum_m h_m e^{-2\pi i f m}$ (re-index $m \mapsto -m$); $h \in \ell^1$ makes the sum absolutely convergent. (Exercise.)

Why: the practical recipe — read off h from the filter, then sum the series to get H (and conversely, recover h as Fourier coefficients of H).

Theorem 11: Filtering preserves stationarity

Let $\{X_t\}$ be stationary and $\mathcal{L}$ an LTI filter with $h \in \ell^1$. Then $\{Y_t\} = \mathcal{L}[\{X_t\}]$ is stationary.

Proof idea: by the convolution theorem $Y_t \stackrel{\text{ms}}{=} \sum_m h_m X_{t-m}$. By Fubini, $\mathbb{E}[Y_t] = \sum_m h_m \mathbb{E}[X_{t-m}] = \mu_X \sum_m h_m$ is t -free; the covariance $\gamma_\tau^{(Y)} = \sum_{m,s} h_m \overline{h_s} \gamma_{\tau-m+s}^{(X)}$ depends only on τ ; and $\gamma_0^{(Y)} \leq \gamma_0^{(X)} \|h\|_1^2 < \infty$, so all is finite.

Why: it licenses speaking of $\mathcal{S}_Y$ at all — only a stationary output has a spectral density.

Theorem 12: Output spectrum: $\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f)$

Let $\{X_t\}$ be stationary with $\gamma^{(X)} \in \ell^1$, and $\mathcal{L}$ an LTI filter with $h \in \ell^1$ and transfer function H . If $\{Y_t\} = \mathcal{L}[\{X_t\}]$ then

$$\mathcal{S}_Y(f) = |H(f)|^2 \mathcal{S}_X(f), \quad f \in \mathbb{R}.$$

Proof idea: from the covariance formula above, $\gamma_\tau^{(Y)} = \sum_u w_u \gamma_{\tau-u}^{(X)}$ where $w_u = \sum_s h_{u+s} \overline{h_s}$ (set $u = m - s$). With $\tilde{h}_t = \overline{h_{-t}}$ one has $w = h * \tilde{h} \in \ell^1$, so the convolution theorem gives $\mathcal{S}_Y = W \mathcal{S}_X$ with $W(f) = H(f) \overline{H(f)} = |H(f)|^2$, since $\tilde{H}(f) = \overline{H(f)}$.

Why: the workhorse identity. It converts *any* filtering question into multiplying the input

spectrum by a gain $|H(f)|^2$ — and, applied to $\Phi(B)$ and $\Theta(B)$, it yields the ARMA spectral density below.

Lemma 13: Transfer function of a polynomial filter $\Phi(B)$

For a degree- p polynomial $\Phi(z) = \sum_{j=0}^p \phi_j z^j$, the filter $\Phi(B)$ is LTI with transfer function

$$H(f) = \Phi(e^{-2\pi i f}), \quad f \in \mathbb{R}.$$

Proof idea: $\Phi(B)$ is a linear combination of the LTI filters B^j , hence LTI, with impulse response $h_m = \phi_m$ for $0 \leq m \leq p$ and 0 otherwise. Then $H(f) = \sum_m h_m e^{-2\pi i f m} = \sum_{j=0}^p \phi_j e^{-2\pi i f j} = \Phi(e^{-2\pi i f})$.

Why: it turns the operator $\Phi(B)$ into a concrete function of f ; the bridge from time-domain difference equations to the spectrum.

Theorem 14: ARMA spectral density

For a stationary ARMA(p, q) process $\Phi(B)X_t = \Theta(B)\varepsilon_t$ with ε_t white noise of variance σ^2 ,

$$\mathcal{S}_X(f) = \sigma^2 \frac{|\Theta(e^{-2\pi i f})|^2}{|\Phi(e^{-2\pi i f})|^2} = \sigma^2 \left| \frac{\sum_{j=0}^q \theta_j e^{-2\pi i j f}}{\sum_{j=0}^p \phi_j e^{-2\pi i j f}} \right|^2.$$

Proof idea: write $\Phi(B)[\{X_t\}] = \Theta(B)[\{\varepsilon_t\}]$. Applying the output-spectrum theorem to each side and the polynomial-filter lemma, $|\Phi(e^{-2\pi i f})|^2 \mathcal{S}_X(f) = |\Theta(e^{-2\pi i f})|^2 \mathcal{S}_\varepsilon(f)$ with $\mathcal{S}_\varepsilon(f) = \sigma^2$; divide, legal because stationarity forbids zeros of Φ on the unit circle, so $\Phi(e^{-2\pi i f}) \neq 0$.

Why: ARMA processes have *no* closed-form ACVS but a clean closed-form spectral density — the practical way to read off their frequency content.